# SCOTT **BATEMAN, PhD**

Professor | Faculty of Computer Science | University of New Brunswick

TABLE OF CONTENTS


SECTION 1 | BIBLIOGRAPHY

SECTION 2 | EDUCATION

SECTION 3 | EMPLOYMENT

SECTION 4 | DISSEMINATION OF KNOWLEDGE

SECTION 5 | RESEARCH, SCHOLARLY OR CREATIVE ACTIVITY

SECTION 6 | SERVICE

SECTION 7 |OTHER RELEVANT INFORMATION


SECTION 1 |
**BIBLIOGRAPHY**


Scott Saunders Bateman
Citizenship Status: Canada

+1 (506) 447-3336(office)

scottb@unb.ca

Web: scottbateman.ca
LinkedIn: ScottBateman
Github: ScottBateman

FACULTY ADDRESS
P.O. Box 4400
550 Windsor Street
Fredericton, NB
E3B 5A3, Canada


## SECTION 2 | **EDUCATION**

**Jan 2008 – Sept 2012 - Doctor of Philosophy**
**Computer Science – Human-Computer Interaction**
University of Saskatchewan, Saskatoon, SK, Canada

Dissertation: *Social Feedback: Learning from Shared Interaction History to Support Information Seeking on the Web*

Advisors: Prof. Carl Gutwin & Prof. Gordon McCalla

**Sept 2005 – Dec 2007 – Master of Science**
**Computer Science – Human-Computer Interaction/Educational Technology**
University of Saskatchewan, Saskatoon, SK, Canada

Thesis: *Collaborative Tagging: folksonomy, metadata, visualization, e-learning*

Advisor: Prof. Gordon McCalla

**Sept 1997 – May 2001 – Bachelor of Science**
**Major: Computer Science, First-class Honours**

University of Prince Edward Island, Charlottetown, PE, Canada

**Sept 1994 – June 1997 – High School Diploma**
Colonel Gray Senior High School, Charlottetown, PE, Canada

**OTHER FORMAL EDUCATION**
**Sept 2006 – May 2007 – Certificate of University Teaching**

University of Saskatchewan, Saskatoon, SK, Canada

**AREA(S) OF ACADEMIC INTEREST:** Human-Computer Interaction, Mixed Reality, Game Design, Serious Games, Human-AI Collaboration, and Computer Supported Cooperative Work.

**ADDITIONAL PROFESSIONAL TRAINING**

2026 **Harassment and Discrimination Policy** | University of New Brunswick Training for managing staff and faculty at UNB.

# SECTION 3 | **EMPLOYMENT**

## 3.1 EMPLOYMENT HISTORY AT UNB

2025 - present **Professor** | Faculty of Computer Science
University of New Brunswick

2024 - present **Research Director of Research Institute on Data Science and Artificial Intelligence**
University of New Brunswick

2023 - present **Director of SPECTRAL Spatial Computing Research Centre** | Faculty of Computer Scie
University of New Brunswick

2019 - 2025 **Associate Professor (tenured 2021)** | Faculty of Computer Science
University of New Brunswick

2015 - 2019 **Assistant Professor (tenure-track)** | Faculty of Computer Science
University of New Brunswick

## 3.2 LIST RELEVANT POSITIONS

2026 **EAISI Visiting Professor** | Eindhoven Artificial Intelligence Systems Institute | Technical University Eindhoven

Provided research seminar and engaged in collaborative research projects.

2025 **Visiting Professor** |Games Technologies Program | Hochschule Bonn-Rhein-Sieg

Provided guest lectures on game user interfaces and mixed reality.

2024 **Visiting Professor** | Département Informatique de la Faculté des Sciences et Technologies
Université de Lille

Provided guest lectures on game design and development and collaborated on research projects.

2023 **Visiting Professor** | School of Computing and Information Systems
Singapore Management University

Ran workshops on HCI research in Mixed Reality and collaborated on research projects.

2015 - 2019 **Adjunct Professor** | Dept. of Computer Science
University of Prince Edward Island

Collaborated on research projects and assisted in the supervision of undergraduate research

2012 - 2015 **Assistant Professor (tenure-track)** | Dept. of Computer Science
University of Prince Edward Island

2013 **Visiting Researcher** | iLab
University of Calgary (12 weeks)

2011 **Research Intern** | Microsoft Research
Redmond, Washington, USA (14 weeks)

2008 **Research Intern** | Centre for Social Software-IBM Research
Cambridge, Mass., USA (14 weeks)

2006 **Research Intern** | Department of Informatics-National College of Ireland
Dublin, Ireland (12 weeks)

2001 – 2005 **Web Developer |** Government of Prince Edward Island and Consulting

### 3.3 LEAVES GRANTED BY UNB OR OTHER UNIVERSITY

2022 **6-month Sabbatical** | Visited Inria - Lille, France and established a new and on-going resea partnership.

### 3.4 DISTINCTIONS, HONOURS, FELLOWSHIPS, SCHOLARSHIPS

2022-2023 **UNB Merit Award** | UNB Merit Awards recognize outstanding contributions to teaching, research and university service by members of the Association of University of New Brunswick Teachers ($5,000).

2022 **Mourou Strickland Mobility Award** | Embassy of France

Award for establishing a research collaboration between UNB, Université de Lille and Inria (~$6,000).

2017 **Harrison McCain Foundation Young Scholars Award**. To support the Building Therapeutic Relaxation Games project ($19,125; 100% of award).

2012 - 2014 **Post-Doctoral Fellowship (NSERC PDF)** | Natural Science and Engineering Research Council (NSERC), to be taken up at Carnegie Mellon University, November 1st, 2012 ($80,000), declined.

2013 **Visiting Researcher Fellowship** | Department of Computer Science, University of Calgary. ($1,000).

2012 **Best Graduate Student** | selected by faculty at the Department of Computer Science, University of Saskatchewan for academic and scholarly achievement, and for contributions made and service to the Department.

**Walter C. Sumner Memorial Fellowship** | Walter C. Sumner Foundation. ($6,000)

**Microsoft Research Student Travel Award** | for CHI 2012, in Austin, Texas. ($500)

**Student Travel Award** | Univ. of Sask., for CHI 2012, in Austin, Texas. ($750)

**Faculty Scholarship** | Univ. of Sask. ($13,700)

2011 **Microsoft Research Student Travel Award** | HCIR 2011, Mountain View, CA. ($500)

**Student Travel Award** | Univ. of Sask., for CHI 2011, Vancouver. ($500)

**Faculty Scholarship** | Univ. of Sask. ($13,700)

**Walter C. Sumner Memorial Fellowship** | Walter C. Sumner Foundation. ($6,000)

2008 – 2011 **National Post Graduate Scholarship (PGS D)** | Natural Sciences and Engineering Research Council (NSERC). ($63,000)

**NSERC Holder's Scholarship** | Univ. of Sask. ($18,000)

**Faculty Scholarship** | Univ. of Sask. ($33,000)

2010 **Student Travel Award** | Univ. of Sask., for CHI 2010, in Atlanta. ($750)

2007 - 2008 **LORNET Research Fellowship** | Univ. of Sask. ($25,000)

2007 **Student Travel Award**, World Wide Web Conference | Banff, Canada. ($500)

2005 - 2007 **College of Graduate Studies and Research Scholarship** | Univ. of Sask. ($15,000)

**Graduate Teaching Fellowship, Univ. of Saskatchewan** | ($15,000)

### 3.5 PAPER AND RESEARCH AWARDS

2026 **Honorable Mention Award** | for the paper Group Conversational Agents: A Review of Designs that Support and Shape Group Interaction, ACM Designing Interactive Systems (DIS '26). Top 5% of submissions.

2025 **Honorable Mention Award** | for the paper Spatially Organized Interfaces for Document Navigation in Mixed Reality, Conference on ACM Spatial User Interfaces (SUI '25). Top 5% of submissions.

2024 **Lasting Impact Award** | This award annually honors one influential paper published at CHI PLAY that has significantly impacted the CHI PLAY community and beyond. In 2024, full papers published from the CHI PLAY conference ten years prior (i.e., 2014) were considered.

Awarded for the paper:

Jared E. Cechanowicz, Carl Gutwin, Scott Bateman, Regan Mandryk, and Ian Stavness. 2014. Improving player balancing in racing games. In Proceedings of the First ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '14).

Association for Computing Machinery, New York, NY, USA, 47–56. https://doi.org/10.1145/2 658 537.2658701

2024 **Honorable Mention Paper Award** | for the paper Incentivizing Intermittent and Frequent Breathing with Idle Game Mechanics, Conf. on Human-Factors in Computing Systems (CHI 2024). Top 5% of submissions.

2011 **Honorable Mention Paper Award** | for the paper Using Target Assistance for Subtly Balancing Competitive Play, Conf. on Human-Factors in Computing (CHI '11). Top 5% of submissions.

2010 **Best Paper Award** | for the paper Useful Junk? The Effects of Visual Embellishments on the Comprehension and Memorability of Charts, Conf. on Human-Factors in Computing (CHI '10). Top 1% of submissions.

2006 **1st Place in the Demo Competition, 2006 LORNET Conference** | for "Collective Intelligence in E-Learning" a demo of the Open Annotation and Tagging System (OATS).

# SECTION 4 | **DISSEMINATION OF KNOWLEDGE**

### 4.1 COURSES TAUGHT

Fall 2026 **CS 3035** – Building User Interfaces – co-taught third year course on the implementation behind user interfaces. 70 students.

Winter 2026 **CS 2053** – Game Development – co-taught third year course on the implementation and design of video games. 67 students.

Fall 2025 **CS 3035** – Building User Interfaces – co-taught third year course on the implementation behind user interfaces. 69 students.

Fall 2024 **CS 3035** – Building User Interfaces – co-taught third year course on the implementation behind user interfaces. 76 students.

Winter 2024 **CS 6999 –** Mixed Reality–senior grad course. Seminar style with a mixed reality-based project. 20 students.

Fall 2023 **CS 3035** – Building User Interfaces –taught third year course on the implementation behind user interfaces. 79 students.

Winter 2023 **CS 6999 –** Mixed Reality–senior grad course. Seminar style with a mixed reality-based project. 20 students.

| | |
|---|---|
| Fall 2022 | **CS 3035** – Building User Interfaces –taught third year course on the implementation behind user interfaces. 69 students. |
| | **CS 6003** – Research Method – introduced new graduate students to research methods and communications. 21 students. |
| Fall 2021 | **CS 3035** – Building User Interfaces –taught third year course on the implementation behind user interfaces. 34 students. |
| | **CS 1003** – Programming and Problem Solving for Engineers – first-year course for non-software engineering, engineering students. Taught Python, programming concepts and problems solving skills. 178 students. |
| Winter 2021 | **CS 4065/6065** – Interactive Human-Centered Systems – taught fourth-year/junior graduate course on introduction to HCI design, evaluation and research. 34 students. |
| | **CS 2053** – Introduction to Game Development – taught game design, development, 3D graphics, and relevant algorithms for games. Used the Unity platform. 44 students. |
| Fall 2020 | **CS 3035** – Building User Interfaces –taught third year course on the implementation behind user interfaces. 67 students. |
| | **CS 6999 –** Mixed Reality Course – newly designed senior grad course. Seminar style with a mixed reality-based project. 7 students. |
| Winter 2020 | **CS 4065/6065** – Interactive Human-Centered Systems – taught fourth-year/junior graduate course on introduction to HCI design, evaluation and research. |
| | **CS 2053** – Introduction to Game Development – taught game design, development, 3D graphics, and relevant algorithms for games. Used the Unity platform. |
| Fall 2019 | **CS 3035** – Building User Interfaces – designed and taught third year course on the implementation behind user interfaces. 48 students. |
| Winter 2019 | **CS 4065/6065** – Interactive Human-Centered Systems – taught fourth-year/junior graduate course on introduction to HCI design, evaluation and research. 29 students (19 CS4065, 10 CS6056). UNB. |
| | **CS 2053** – Introduction to Game Development – taught game design, development, 3D graphics, and relevant algorithms for games. Used the Unity platform. 38 students. UNB. |
| Fall 2018 | **CS 1073** – Introduction to Computer Science I – taught Java, problem solving, program design. 44 students. UNB. |
| Winter 2018 | **CS 2053** – Introduction to Game Development – taught game design, development, 3D graphics, and relevant algorithms for games. Used the Unity platform. 38 students. UNB. |

**CS 4065/6065** – Interactive Human-Centered Systems – taught fourth-year/junior graduate course on introduction to HCI design, evaluation and research. 26 students (18 CS4065, 8 CS6056). UNB.

Fall 2017

**CS 1073** – Introduction to Computer Science I – taught Java, problem solving, program design. 44 students. UNB.

Winter 2017

**CS 1083** - Introduction to Computer Science II – taught Java, problem solving, program design. 39 students. UNB. Course director (coordinated with two other course sections on a common schedule, assignments, labs and exams).

**CS 4065/6065** – Interactive Human-Centered Systems – taught fourth year/junior graduate course on introduction to HCI design, evaluation and research. 18 students. UNB.

Fall 2016

**CS 1073** – Introduction to Computer Science I – taught Java, problem solving, program design. 39 students. UNB.

**CS 3025** – Human-Computer Interaction – taught introductory concepts in interface design, requirements and needs finding, and interface evaluation. 41 students. UNB.

**CS 2999** – Problem Solving for Programmers – supervised independent study course on advanced problem-solving techniques. 7 students. UNB.

Winter 2016

**CS 1083** – Introduction to Computer Science II – taught Java, problem solving, program design. 38 students. UNB.

**CS 2999** – Problem Solving for Programmers – supervised independent study course on advanced problem-solving techniques. 8 students. UNB.

**CS 4995/6905** – Interactive Human-Centered Systems – taught fourth year/junior graduate course on introduction to HCI design, evaluation and research. 33 students. UNB.

Fall 2015

**CS 1083** – Introduction to Computer Science II – taught Java, problem solving, program design. 24 students. UNB.

Winter 2015

**CS 151** – Introduction to Computer Science I – taught Java, problem solving, program design. 34 students. UPEI.

**CS 482** – Software Systems Development Project – Supervise the development of capstone group projects, ensure that software engineering process and principles are correctly applied, advise on logistical challenges and conflict resolution. 10 students. UPEI.

**CS 483** – Video Game Programming Project – Supervise the development of capstone game for video game specialization, ensure that software engineering process and principles are correctly applied, communicate and work with industry advisors. 1 student. UPEI.

| | |
|---|---|
| Fall 2014 | **CS 261** – Data Structures – Taught the design and implementation of data structures, sorting algorithms, and basic algorithm analysis. Overall student rating: 4.3/5, 26 students. UPEI. |
| | **CS 495** – Data Science – Designed the curriculum for this new course, without the benefit of a textbook. Taught computational statistics, machine learning, and visualization. Overall student rating: 4.8/5, 17 students. UPEI. |
| Winter 2014 | **CS 151** – Introduction to Computer Science I – taught Java, problem solving, program design. Overall student rating: 4.2/5, 28 students. UPEI. |
| | **CS 152** – Introduction to Computer Science II – taught Java, problem solving, program design. Overall student rating: 4.5/5, 32 students. UPEI. |
| Fall 2014 | **CS 151** – Introduction to Computer Science I – taught Java, problem solving, program design. Overall student rating: 4.2/5, 35 students UPEI. |
| | **CS 321** – Human-Computer Interaction – taught need finding, interface design and evaluation. Overall student rating: 4.2/5, 15 students. UPEI. |
| Winter 2013 | **CS 151** – Introduction to Computer Science I – taught Java, problem solving, program design. Overall student rating: 4.3/5, 25 students. UPEI. |

### 4.2 DEVELOPMENT OF NEW COURSES

CS 6015 (2020) – Mixed Reality (Graduate level)

CS 3035 (2019) – Building User Interfaces

CS 4065/6065 (2016) – Interactive Human-Centered Systems

### 4.3 UPDATING OF EXISTING COURSES

CS 1003 (2021) – Programming and Problem Solving for Engineers (in collaboration with a student assistant)

CS 2053 (2018) – Video Game Development

### 4.4 EFFECTIVE USE OR DEVELOPMENT OF TEACHING AIDS & TRAINING

2020 – Certificate of Participation in UNB CETL "How to Adapt Your Course for the Fall: Everything you need to know."

### 4.5 INVOLVEMENT IN CURRICULUM DEVELOPMENT

In 2016, I established a shared and co-organized a shared repository of HCI curriculum, course materials and educational activities for Canadian HCI Faculty. The repository has been used by over 40 members.

### 4.6 ORGANIZATION OF FIELD SCHOOLS, LABS OR PRACTICA

NextGenUX 2018: Co-organizer for NextGenUX. A two-day tutorial and workshop event to introduce HCI research and practice to New Brunswick industry and government. Fredericton, NB. April 2018. The 20-day event attracted over 110 attendees; I saw presentations from 7 international HCI research leaders.

### 4.7 a) DIRECTION OF RESEARCH ON UNDERGRADUATE OR GRADUATE STUDIES

#### UNDERGRADUATE RESEARCH ASSISTANTS

| | |
|---|---|
| Summer 2026 | Renata Lara Lopez (Mitacs Research Intern - Universidad Autónoma de San Luis Potosí). Supervised research on foot-based gesture input. |
| | Isabella Lopez Ibarra (Mitacs Research Intern - Universidad Autónoma de Occidente). Supervised research on transfer functions in mixed reality. |
| | Jashendeep Singh Sidhu (UNB) student working on computer vision automated fish health classification system (with Cooke Aquaculture). |
| | Erin Macneil (UNB) student working on improving memory for RAG-based LLMs. |
| Summer 2024 | Amy Cui (Queen's University) student working on mixed reality tutorial systems. |
| | Jakob Ashfield (UNB) student working on novel wearable input device for mixed reality. |
| | Alexander Groom (UNB) student working on collaboration in mixed reality. |
| Summer 2023 | Nolan Kisser (Carleton University) student working on remote mixed reality support systems Q&A with collaborator Kognitiv Spark. |
| | Jakob Ashfield (UNB) student working on remote mixed reality support systems Q&A with collaborator Kognitiv Spark. |
| Summer 2022 | Book Sadprasid (UNB) NSERC USRA student worked on game to breathing exercises, and technology to support collaboration in mixed reality. |
| | Heena Thadani (Mody Institute of Technology, Lakshmangarh, India) MITACS Globalink student worked on a platform for collaboration in mixed-reality. |
| | Deep Fumatiya (Veermata Jijabai Technological Institute, Mumbai, India) MITACS Globalink student working on processing and visualizing large 3D datasets. |
| Summer 2021 | Book Sadprasid (UNB) NSERC USRA student working on game to support chest physiotherapy exercises. |
| Summer 2020 | Book Sadprasid (UNB) NSERC USRA student working on qualitative analysis of novel computer interactions. |
| Summer 2019 | Noor Ali (UNB) for research on developing an augmented reality-based game to connect families at a distance. |

| | |
|---|---|
| | Nikita Baliesnyi (Taras Shevchenko National University of Kyiv), MITACS Globalink student creating a web toolkit for multi-display environments. |
| | Connor Wilson (UNB), NSERC USRA student, worked on creating sensory illusions in Augmented Reality. |
| Winter 2019 | Nathan Sonier (UNB) for research on developing an in-home, health visit video conferencing system. |
| Summer 2018 | Wesley Finck (UNB) NSERC USRA student, research on developing a virtual prosthesis trainer. Co-supervised (with Erik Scheme). |
| Summer 2017 | Ben Rombaut (UNB) for his research on Developing Therapeutic Gait Feedback for a Smart-Cane. Co-supervised (with Erik Scheme) |
| | Nathaniel Brewer (UNB) for his research on Developing an experimental platform for evaluating spatial interfaces on mobile devices (Aug. 2017-Sept. 2017). |
| Summer 2016 | Carly Smith (UNB) for her research on Developing a Training Game for Upper Limb Prosthesis Users. Co-supervised (with Erik Scheme). |
| | Alex Watson (UNB) for his research on Developing a Training Game for Upper Limb Prosthesis Users. Co-supervised (with Erik Scheme). |
| | Alex Kienzle (UNB) for his research on Visualizing Predictive Analytics Data. |
| | David Hanna (UPEI) for his research on a multi-display framework for games. |
| Winter 2016 | Danielle Brockway (UNB), in the development for a research lab administrative system. |
| Summer 2015 | Courtney Horrocks (UPEI) NSERC USRA student, research on Communicating and Visualizing Correlation Data for the General Public. |
| Summer 2014 | Igor Margues da Silva (Brazil) in his research and development of collaborative integrated development environments. Co-supervised with Carl Gutwin, student hosted at the University of Saskatchewan. Co-supervised (with Erik Scheme). |
| | Nikita Volodin (UPEI) in the development of multi-surface application framework. |
| | Yixin Wang (UPEI) in the development of a crowd-sourcing experiment system. |
| | Harvey Xia (UPEI) in the development of research lab support applications. |
| | Nguyen Cong Van (UPEI) in the development of collaborative video conferencing systems. |
| | James Martin (UPEI) in the development of a zoomable web interface. |
| | Nikita Volodin (UPEI) in the development of multi-surface application framework. |
| | Ivy Truong (Dalhousie University) in the development of an experimental system to test new pointing techniques. |

| | |
|---|---|
| Summer 2013 | Camila Caraujo (Univ. Federal de Minas Gerais, Brazil) in the development of social mobile search app. |
| | Maria Clara Pestana (Universidade Federal da Paraíba, Brazil) in the development of mobile map-based search app. |
| | Lucas Rosa (Instituto Federal Catarinense, Brazil) in the development of a web-based IDE for large-scale capture of programming behaviour. |

**BACHELOR (HONOURS THESIS)**

| | |
|---|---|
| 2025 | Maryna Myronenko (UNB) – Supervising honours thesis on understanding the effect of Cost of Effort on Usability Experience. Expected May 2025. |
| | Promise Abel-Adegbite (UNB) – Supervised honours thesis on a comparison of computer vision tracking technology for capturing biomechanics of sprinters. Expected May 2025. |
| 2022 | Kolton Gagnon (UNB) - Supervised honours thesis on a Web Application Framework for Multi-surface Applications. Completed May 2022. |
| | Book Sadprasid (UNB) - Supervised honours thesis on a Therapeutic Game of Cystic Fibrosis. Completed May 2022. |
| 2018 | Thierry Peltier (UNB) – Supervised honours thesis on Spatial Memory Interfaces for Mobile Devices. Completed May 2018. |
| 2016 | Stanley Petley (UNB) – Co-supervised (with Paul Cook, UNB) honours thesis on Visualizing Personal Affect from Social Media Streams. Completed May 2016. |
| | Nikita Volodin (UPEI) - Supervising honours thesis project on Analyzing and Identifying Successful Behaviour in MOBA Games. Completed May 2016. |
| | Jason Wuertz (UPEI) - Supervising honours thesis project on Understanding the Use of Interface Features for Coordination and Collaboration in MOBA Games. Completed May 2016. |

**MASTERS THESIS (IN PROGRESS)**

| | |
|---|---|
| 2026-2028 | Shantu Latke – MCS Student. Co-supervised with Aaron Tabor. Research TBD. |
| 2025-2027 | Charles Ewaifoh Ejedawe – MCS Student. Co-supervised with Aaron Tabor. Supervising research on detecting activities using heart-rate variability signal classification. |
| | Nashmia Riaz – MCS (Report) Student. Supervising research on designing the effect of player selected challenges on player experience in video games. |

**MASTER'S THESIS COMPLETED**

| | |
|---|---|
| 2023-2026 | Taylor Short – MCS Student. Supervising research on Mixed Reality games to connect older adults with family members. Completed August 2026. |
| | Fahimeh Talouee – MCS Student. Supervising research on the effects of embodiment in mixed reality. Completed August 2026. |

| | |
|---|---|
| 2023-2025 | Amirhossein Azimyzadeh – MCS Student. Supervising research on improving desktop interactions in mixed reality. Completion November 2025. |
| 2022-2025 | Cole Campbell – MCS Student. Supervising research on MR tutorials for nursing education. Completion August 2025. |
| | Colin Brett – MCS Student. Supervising research on displays alternatives in mixed reality. Completion August 2025. |
| 2022-2024 | Oyindolapo Komolafe – MCS Student. Supervising research on the use of augmented reality in construction and its effect on situational awareness. Completed July 2024. |
| 2021-2023 | Joshua Roberts – MCS Student. Supervising research on the effect of music on play experience. Completed 2023. |
| | Nick Balcomb – MCS Student. Supervising research on the effect of controller representations on player experience. Completed 2023. |
| | Alexander LeBlanc – MCS Student. Supervising research on asymmetric game play. Expected completion Winter 2023. |
| 2019-2021 | Samridhi Pargal – MCS Student. Supervised research on the Simple Mobile Augmented Reality Toolkit. Completed Dec 2021. |
| | Connor Wilson – MCS Student. Supervised research on how visual design affects perception of credibility of social media new stories. Completed August 2021. |
| | Isayah Vidito – MCS Student. Co-supervised (with Dawn MacIsaac, Computer Science, UNB) research on remote guidance of therapeutic exercise in mixed reality. Completed April 2021. |
| | Benjamin MacPherson – MCS Student. Co-supervised (with Michael Fleming, Computer Science, UNB) research on intelligent adaptation in games. Complete June 2021. |
| | Alex Kienzle – MCS Student. Supervised research on augmented reality-based collaboration. Completed May 2021. |
| 2018 –2020 | Reyhan Pradantyo – MCS Student. Supervised research on modeling the design of antagonists in video games. Completed September 2020. |
| | Cassandra Folkins – Masters of Applied Health Research. Co-supervised with Emily Read (Nursing, UNB) research on serious games for promoting careers in nursing homes. Completed May 2020. |
| 2017 –2019 | Vinod Asokan – MCS Student. Supervised research on assistance for pointing-based interactions in augmented reality. Completed fall 2019. |
| | Ian Smith – MCS Student. Co-supervised (with Erik Scheme, Institute of Biomedical Engineering, UNB) research on visual feedback during therapeutic activities. Completed summer 2019. |

| | |
|---|---|
| 2016 –2018 | Jason Wuertz– MCS Student. Supervised research on game awareness widgets. ***Awarded Best Thesis for the Faculty of Computer Science.*** |
| | Manasi Shah – MCS Student. research on spatial interfaces for mobile devices. Completed fall 2018. (masters report-based student) |
| | Jawad Jandali – MCS Student. Co-supervised (with Michael Fleming, UNB) modeling abilities in core game tasks to support game balancing. |
| 2016 - 2017 | Aaron Tabor – MCS Student. Co-supervised (with Erik Scheme, UNB) project on training games for upper-limb prosthetics. Completed summer 2017. ***Awarded Best Thesis for the Faculty of Computer Science***. |

**PhD (IN-PROGRESS)**

| | |
|---|---|
| 2025-present | Cole Campbell – PhD in Computer Science. Dissertation on The Design of AI Tools for University Instructors. Expected 2029. |
| | Colin Brett – PhD in Computer Science. Dissertation on the The Design of Tools to Support Digital Minimalism. Expected 2029. |
| 2021-2027 | Ethan Eddy – PhD in Electrical Engineering, co-supervised (with Erik Scheme, UNB) project on EMG control of everyday interfaces and EMS guidance. Expected April 2027. |

**PhD (COMPLETED)**

| | |
|---|---|
| 2019-2026 | Ian Smith – PhD in Computer Science, co-supervised (with Erik Scheme, UNB) thesis on sport training games. Completed August 2026. |
| 2018-2024 | Aaron Tabor – PhD in Computer Science, co-supervised (with Erik Scheme, UNB) thesis on technology to support and promote breathing exercises. Completed August 2024. |

**POST-DOCTORAL FELLOWS**

| | |
|---|---|
| 2025 | Stuart Thiel, PhD industrial post doc working on game development technologies in collaboration with IGT. |
| 2022-2023 | Alec McLellan, PhD conducting research on building mapping tools for archaeological software. |
| 2020-2022 | Cora Woolsey, PhD conducting research on community building in the maker movement, and technology support tools for archaeological field work. Started spin-off company ArchaeoSoft that creates technology for archaeological field work, based on work from post-doctoral researcher. |

## UNB GRADUATE COMMITTEE MEMBERSHIP

| | |
|---|---|
| 2028 | PhD Interdisciplinary Studies. Committee member for Emily Vessey. Intersectionality and future autonomous systems. |
| 2026 | PhD in Computer Science, Computer Science Committee member for Mozzamel Hossain |
| 2025 | PhD in Civil Engineering, External Committee member for Kamyab Aghajamali, UNB. PhD Dissertation: A Decision Support System for Optimized Off-Site Construction Planning Using a Pull-Based Simulation Approach. October 2025. |
| 2023 | MSc. in Engineering, External Committee member for Mahsa Rezvani, UNB. Masters Thesis: Mixed Reality in Precast Concrete Construction: A Case Study of Quality Control Inspection Tasks. June 2023. |
| 2021 | MSc. in Engineering, External Committee member for Ishtar Al-Tahir, UNB. Masters Thesis: Myoelectric Co-Adaptation: Stochastic and Chaotic, yet Describable. March 2021. |
| | MCS Committee member for Samin Fakharian, Faculty of Computer Science. MCS Thesis: Contextualized Embeddings Encode Knowledge of English Verb-Noun Combination Idiomaticity. April 2021. |
| 2020 | MA Committee (Interdisciplinary Studies) member for Jeff Mundee, Department of Media Arts and Culture, UNB. Masters Thesis: Conflict in an Online Game Mod Community. |
| 2019 | MSc. in Engineering, External Committee member for Zhihao Yu, UNB. Masters Thesis: FPGA-based IGBT Switching Spike Removal. |
| | PhD Committee member and chair for Eltaher M. El-Shanta, Faculty of Computer Science, UNB. PhD Dissertation title: Software Patterns as Services. (2015-2019) |
| | MCS Committee Chair for Alex Watson, MCS Thesis: Scalable Unified Data Analytics. |
| | MSc Committee member for Deanna Musgrave, Interdisciplinary Studies, MSc. Thesis title: Mindful Multimedia: Connecting Crossmodal Interactions and Aesthetic Experiences in Visual Music. |
| | MSc. In Engineering External Committee member for Neil Chambers, UNB. Masters Thesis: A Dynamic Approach to Balance Assessment using Self-Induced Peturbations. |
| 2018 | MCS Committee chair for Dmitry Shcherbakov, Faculty of Computer Science, UNB. MCS Thesis title: Application of Reinforcement Learning to Autonomous Aircraft Control in Partially Observable Environments. |
| 2017 | PhD Committee member for Christina Dore, Interdisciplinary Studies, UNB. PhD Dissertation title: Assessing and addressing the risk for burnout among hemodialysis nurses working in Quebec. |
| | PhD Examining Committee Member and External Defence Chair for Katerina Biron, Electrical Engineering, UNB. PhD Dissertation title: Towards Improvement of the |

| | |
|---|---|
| | Dynamic Performance of Myoelectric Pattern Recognition for Improved Real-Time Usability. |
| | MCS Committee chair for Vaibhavi Kalgutkar, Faculty of Computer Science, UNB. MCS Thesis title: Android authorship attribution through string analysis. |
| 2016 | M.Sc. in Engineering, Committee member for Gillian Phillips, Faculty of Engineering, UNB. MSc. Thesis title: Pairwise Attribute Noise Detection Algorithm for Detecting Noise in Surface Electromyography Recordings. |

### 4.7 b) EXTERNAL EXAMINER FOR Ph.D THESIS

| | |
|---|---|
| 2024 | External Examiner for Yuan Chan, PhD in Computer Science, at Université de Lille (France) and University of Waterloo (co-tutelle). September 2024. |
| 2023 | External Examiner for Adrian Schneider, PhD in Computer Science, Queen's University. Personalized dynamic modelling of spatiotemporal aiming ability for systematic player balancing. March 2023. |
| 2021 | External Examiner for Milad Soroush, PhD in Management Sciences, University of Waterloo. The Impacts of Gameful and Interactive Technologies on Hindering or Promoting Self-regulation. March 2021. |

### 4.7 c) EXTERNAL EXAMINER FOR MASTER'S THESIS

| | |
|---|---|
| 2017 | External Examiner for Rui Pan, M.Sc. in Interactive Arts and Technology, Simon Fraser University. Thesis title: First-Person Video Chat for Couples at a Distance. July 2017. |
| | External Examiner for Radhika Gopi, M.Sc. in Computer Science, University of Regina. Thesis title: Twist: A Mobile Approach for Searching and Exploring within Twitter. May 2017. |

### 4.8 PUBLICATION OR PRODUCTION OF BOOKS, ARTICLES, FILMS, ETC. THAT ADVANCE TEACHING EFFECTIVENESS

N/A

### 4.9 AWARD HONOURS FOR TEACHING

N/A

### 4.10 ANY OTHER EQUIVALENT ACCOMPLISHMENTS

**SUPERVISION OF FULL TIME STAFF**

| | |
|---|---|
| 2025 – present | Amirhossein Azimyzadeh, HCI Research Developer (full time) supporting projects of SPECTRAL. |
| 2024 – present | Aaloak Jaswal, Innovation Director (full time) supporting projects of RIDSAI. |
| 2024 - 2025 | Passionate Ncube, Project Coordinator (full time) supporting projects of RIDSAI. |
| 2023 - present | Vikki Babic, Innovation Director (full time) supporting projects of SPECTRAL. |
| | Jeff Mundee, HCI Research Developer (full time) supporting projects of SPECTRAL. |
| 2023-2024 | Book Sadprasid, HCI Research Developer (full time) supporting projects of SPECTRAL. |
| 2022 | Jeff Mundee research assistant for An AI Approach to Electronic Gaming Machine (EGM) Game Modeling: Reward Function. In collaboration with IGT Canada Solutions UL ArchaeoSoft, Inc (Moncton, NB). |
| | David Milley research assistant for Understanding the Usability of Digital Field notes tool for archaeologists. In collaboration with ArchaeoSoft, Inc (Fredericton, NB). |
| | Thomas Boles research assistant for Understanding the Usability of Digital Field notes tool for archaeologists. In collaboration with ArchaeoSoft, Inc (Fredericton, NB). |
| | Andrew Vissers developer for a Digital Field notes tool for archaeologists. In collaboration with ArchaeoSoft, Inc (Fredericton, NB). |
| | Mattias Schroer developer for a Digital Field notes tool for archaeologists. In collaboration with ArchaeoSoft, Inc (Fredericton, NB). |
| | Jacob Beaumont developer for a Digital Field notes tool for archaeologists. In collaboration with ArchaeoSoft, Inc (Fredericton, NB). |
| 2020 | Siddharth Yadav in Development of a Mobile Interface for the presentation of health data to non-expert stake holders in collaboration with VeroSource, Inc (Fredericton, NB). |
| | Khanjan Barad in Development of a Mobile Interface for the presentation of health data to non-expert stake holders in collaboration with VeroSource, Inc (Fredericton, NB). |
| 2016 | Aakash Shukla in the research and development of the Nursing Central mobile app (Jan. 2016-May 2016) at UNB. |
| 2014 - 2015 | Nolan Phillips, BSc in the development of a crowdsourcing solution to identify sales leads. In collaboration with nGauge, Inc., supported by NSERC Engage grant. (Sept. 2014 – Nov. 2015) at UPEI. |
| | Nguyen Cong Van, BSc in the development of a video communication system in collaboration with XMG, Inc (Sept. 2014-Jan. 2015) at UPEI. |
| | Peter Workman, BSc in the development of a virtual reality desktop computing system (Mar. 2014-July 2014). |

# SECTION 5 | **RESEARCH, SCHOLARLY OR CREATIVE ACTIVITY**

## 5.1 ARTICLES PUBLISHED OR ACCEPTED FOR PUBLICATION IN REFEREED JOURNALS

| | |
|---|---|
| CHI PLAY 2026 | Ian Smith, Erik J Scheme, and Scott Bateman. 2026. Comparing Scoring Mechanics for Encouraging Technique Adoption in a Sport Training Game. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), 1–19. https://doi.org/10.1145/3772318.3791286 |
| ML Health 2025 | Ethan Eddy, Evan Campbell, Christian Morrell, Heather Williams, Scott Bateman, Erik Scheme.Raising the Standard: An Open-Source Benchmarking Platform and Data Repository to Accelerate Myoelectric Control Research. Machine Learning: Health, vol. 1, no. 1, 2025, DOI: 10.1088/3049-477X/ade549 |
| F&T-HCI 2025 | Ian Smith, Erik Scheme, Scott Bateman. A Framework for Interactive Sport Training Technology. Foundations and Trends® in Human-Computer Interaction 19 (1). 111 pages. http://dx.doi.org/10.1561/1100000096 |
| JNE 2025 | Ethan Eddy, Evan Campbell, Scott Bateman and Erik Scheme. EMG-based wake gestures eliminate false activations during out-of-set activities of daily living: an online myoelectric control study. Journal of Neural Engineering, 22, 1. 13 pages. https://doi.org/10.1088/1741-2552/ada4df |
| CHI PLAY 2024 | Ian Smith, Erik Scheme, and Scott Bateman. Designing a Technique-Oriented Sport Training Game for Motivating a Change in Running Technique. Proc. ACM Hum.-Comput. Interact., CHI PLAY, November 2025, 28 pages. https://doi.org/10.1145/3677064 |
| FBIO 2024 | Ethan Eddy, Evan Campbell, Scott Bateman, Ethan Scheme. Big data in myoelectric control: large multi-user models enable robust zero-shot EMG-based discrete gesture recognition. Frontiers in Bioengineering and Biotechnology, 12. 24 pages. https://doi.org/10.3389/fbioe.2024.1463377 |
| TNSRE 2024 | Evan Campbell, Ethan Eddy, Xavier Isabel, Scott Bateman, Benoit Gosselin, Ulysse Côté-Allard, Erik Scheme. Screen Guided Training Does Not Capture Goal-Oriented Behaviors: Learning Myoelectric Control Mappings From Scratch Using Context Informed Incremental Learning. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 33. 2024. 332-342. https://doi.org/10.1109/TNSRE.2024.3518059 |
| JNE 2024A | Robyn Meredith, Ethan Eddy, Scott Bateman and Erik Scheme. Comparing online wrist and forearm EMG-based control using a rhythm game-inspired evaluation environment. Journal of Neural Engineering, 21, 3. August 2024. https://doi.org/10.1088/1741-2552/ad692e |
| JNE 2024B | Ethan Eddy, Evan Campbell, Scott Bateman, and Erik Scheme. Understanding the influence of confounding factors in myoelectric control for discrete posture recognition. Journal of |

| | |
|---|---|
| LETTERS 2024 | Evan Campbell, Ethan Eddy, Ulysse Côté-Allard, Scott Bateman, Erik Scheme. Discrete Gesture Recognition Using Multimodal PPG, IMU, and Single-Channel EMG Recorded at the Wrist. IEEE Sensors Letters, 8, 9. 4 pages. https://doi.org/10.1109/LSENS.2024.3447240 |
| JNER 2024 | Evan Campbell, Ethan Eddy, Scott Bateman, Ulysse Côté-Allard, and Erik Scheme. Context-informed incremental learning improves both the performance and resilience of myoelectric control. Journal of NeuroEngineering and Rehabilitation, 21, 70 (2024). 17 pages. https://doi.org/10.1186/s12 984-024-01 355-4 |
| CHI PLAY 2023 | Nicholas Balcomb, Max V. Birk, and Scott Bateman. 2023. The Effects of Hand Representation on Experience and Performance for 3D Interactions in Virtual Reality Games. Proc. ACM Hum.-Comput. Interact. 7, CHI PLAY, Article 420 (November 2023), 28 pages. https://doi.org/10.1145/3 611 066 |
| EICS 2023 | Scott Bateman, Carl Gutwin, Hamid Mansoor, Miguel Nacenta, Michael van der Kamp, Mykyta Baliesnyi, Kolton Gagnon, and Jesse Rollheiser. 2023. WAMS: A Flexible API for Visual Workspaces Across Multiple Surfaces. Proc. ACM Hum.-Comput. Interact. 7, EICS, Article 184 (June 2023), 40 pages. https://doi.org/10.1145/3 593 236 |
| IEEE ACCESS 2023 | Ethan Eddy, Evan Campbell, Angkoon Phinyomark, Scott Bateman and Erik Scheme. LibEMG: An Open Source Library to Facilitate the Exploration of Myoelectric Control, in IEEE Access, vol. 11, pp. 87 380-87 397, 2023, doi: https://doi.org/10.1109/ACCESS.2023.3304544 |
| FRONTIERS 2022 | Aaron Tabor, Scott Bateman, Erik Scheme, m.c. schraefel. Comparing heart rate variability biofeedback and simple paced breathing to inform the design of guided breathing technologies. Frontiers in Computer Science, vol. 4. https://www.frontiersin.org/articles/10.3389/fcomp.2022.926649 |
| CHI PLAY 2021 | Tim Naglé, Scott Bateman, Max V. Birk. Pathfinder: The Behavioural and Motivational Effects of Collectibles in Gamified Software Training. In the Proceedings of the ACM on Human Computer Interaction, vol 5. (CHI PLAY). 1-23. https://doi.org/10.1145/3 474 691 |
| FRONTIERS 2021 | Reyhan Pradantyo, Max V. Birk, Scott Bateman. How the Visual Design of Video Game Antagonists Affects Perception of Morality. Frontiers in Computer Science. https://doi.org/10.3389/fcomp.2021.531713 |

| | |
|---|---|
| FRONTIERS 2020 | Jawad Jandali Refai, Scott Bateman, Michael W. Fleming. External Assistance Techniques that Target Core Game Tasks for Balancing Game Difficulty. Frontiers in Computer Science, Issue 2, 17 pages. https://doi.org/10.3389/fcomp.2020.00017 |
| IJHCS 2019 | Barrett Ens, Joel Lanir, Anthony Tang, Scott Bateman, Gun A Lee, Thammathip Piumsomboon, Mark Billinghurst. Revisiting Collaboration through Mixed Reality: The Evolution of Groupware-Supplemental Visualization. International Journal of Human-Computer Studies. In press. Available online May 25, 2019. https://doi.org/10.1016/j.ijhcs.2019.05.011 |
| TNSRE 2018 | Aaron Tabor, Scott Bateman, Erik Scheme. Evaluation of Myoelectric Control Learning using Multi-Session Game-Based Training. IEEE Transactions on Neural Systems and Rehabilitation Engineering. Accepted, to be published. https://doi.org/10.1109/TNSRE.2018.2855561 |
| TVCG 2015 | Dandan Huang, Melanie Tory, Bon Adriel Aseniero, Lyn Bartram, Scott Bateman, Sheelagh Carpendale, Anthony Tang, Robert Woodbury. Personal Visualization and Personal Visual Analytics. IEEE Transactions on Visualization and Computer Graphics. Volume 21, Issue 3, March, 2015. DOI: http://dx.doi.org/10.1109/TVCG.2014.2359887 |
| IJHCS 2013 | Scott Bateman, Regan L. Mandryk, Carl Gutwin, Robert Xiao. Analysis and Comparison of Target Assistance Techniques for Relative Ray-Cast Pointing. International Journal of Human-Computer Studies. Volume 71, Issue 5, May 2013, 511-532. http://dx.doi.org/10.1016/j.ijhcs.2012.12.006 |
| ILE 2009 | Jelena Jovanovic, Dragan Gasevic, Carlo Torniai, Scott Bateman, Marek Hatala. The Social Semantic Web in Intelligent Learning Environments-State of the Art and Future Challenges. Interactive Learning Environments Journal, Volume 17, Number 4. 2009. pp. 273–308. |

### 5.2 REFEREED CONFERENCE PROCEEDINGS[1]

| | |
|---|---|
| CHI 2026A | Ethan Eddy, Evan Campbell, Erik J Scheme, Scott Bateman, and Géry Casiez. 2026. TFTune: Creation and Personalization of Pointing Transfer Functions Using Reinforcement Learning. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13--17, 2026, Barcelona, Spain.* ACM, New York, NY, USA. 13 Pages. https://doi.org/10.1145/3772318.3790291 |

[1] In my field of Human Computer Interaction, traditionally, conferences were the preferred and more prestigious publication venues. However, over the past several years, many ACM HCI Conferences have transitioned to journal publication models (e.g., CHI PLAY and EICS), while maintaining the in-person conferences and paper presentations of work is still expected. The ACM CHI Conference is still considered the premier publication venue in HCI, but has not and likely will not transition to a journal model because of the size and traditions of the community.

CHI 2026B — Ethan Eddy, Evan Campbell, Erik J Scheme, and Scott Bateman. 2026. Open, Accurate, and Calibration-Free Muscle-Computer Interfaces. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13--17, 2026, Barcelona, Spain.* ACM, New York, NY, USA. 14 Pages.

CHI 2026C — Ian Smith, Erik J Scheme, and Scott Bateman. 2026. Comparing Scoring Mechanics for Encouraging Technique Adoption in a Sport Training Game. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13--17, 2026, Barcelona, Spain.* ACM, New York, NY, USA. 28 Pages.

GI 2026A — Colin A. Brett, Ernst Kruijff, André Hinkenjann, and Scott Bateman. Flow and Long-Form Reading in Mixed Reality. In Proceedings of Graphics Interface (GI '26). 12 pages.

GI 2026B — Amir Azimyzadeh, Colin Brett, Fahimeh Tolouee, Cole K. Campbell, Scott Bateman. Designing Contextual Toolbars. In Proceedings of Graphics Interface (GI '26). 12 pages.

ICIR 2026 — Alireza Rahimi, Amirhossein Azimyzadeh, Jeff Mundee, Kevin Wilcox, Shadnam Jabari, Scott Bateman, Ian Church, Hung Cao. Towards a Scalable Digital Twin Framework Architecture for Port Resilience: Leveraging IoT, Edge AI, and Data-Driven Streaming for Seamless Physical-Virtual Convergence. *2026 IEEE 5th International Conference on Intelligent Reality (ICIR)* 1–6. 10.1109/ICIR70631.2026.11628305

DIS 2026 — ShunYi Yeo, Tianyi Zhang, Scott Bateman, Gary Hsieh, Young-Ho Kim, Simon Tangi Perrault, Jiannan Li, and Anthony Tang. Group Conversational Agents: A Review of Designs that Support and Shape Group Interaction. In Proceedings of the 2026 Designing Interactive Systems Conference (DIS '26), 2765–2779. https://doi.org/10.1145/3800645.3812934 **Honorable Mention Paper Award (top 5% of submissions).**

SUI 2025A — Martin Pluisch, Scott Bateman, André Hinkenjann, and Ernst Kruijff. 2025. Extended Workspace: Techniques for Interaction with Off-Screen Objects in Augmented Reality. In Proceedings of In ACM Spatial User Interfaces (SUI '25). 12 pages.

SUI 2025B — Colin A. Brett and Scott Bateman. 2025. Spatially Organized Interfaces for Document Navigation in Mixed Reality. In Proceedings of In ACM Spatial User Interfaces (SUI '25). 12 pages. **Honorable Mention Paper Award (top 5% of submissions).**

SUI 2024A — Book Sadprasid, Carl Gutwin, and Scott Bateman. 2024. Improving Video Navigation for Spatial Task Tutorials by Spatially Segmenting and Situating How-To Videos. In Proceedings of the 2024 ACM Symposium on Spatial User Interaction (SUI '24). Article 14, 13 pages. https://doi.org/10.1145/3677386.3682103

SUI 2024B — Ian Smith, Erik J Scheme, and Scott Bateman. 2024. Mapping Real World Locomotion Speed to the Virtual World in Large Field of View Virtual Environments. In Proceedings of the 2024 ACM Symposium on Spatial User Interaction (SUI '24), Article 21, 12 pages. https://doi.org/10.1145/3677386.3682077

SUI 2024C — Oyindolapo Olabisi Komolafe, Colin A. Brett, Pouya Pournasir, Lloyd Waugh, Zhen Lei, Daniel J Rea, and Scott Bateman. 2024. The Effect of Augmented Reality on Performance, Task

| | |
|---|---|
| | Loading, and Situational Awareness in Construction Inspection Tasks. In Proceedings of the 2024 ACM Symposium on Spatial User Interaction (SUI '24). Article 10, 12 pages. https://doi.org/10.1145/3677386.3682082 |
| CHI 2024A | Book Sadprasid, Anne Mei, Alex Mariakikis, Scott Bateman and Fanny Chevalier. Incentivizing Intermittent and Frequent Breathing with Idle Game Mechanics. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2024). Honolulu, USA. 2024. Association for Computing Machinery, New York, NY, USA, Article 714, 1–17. https://doi.org/10.1145/3613904.3642430 **Honorable Mention Paper Award (top 5% of submissions).** |
| CHI 2024B | Florian 'Floyd' Mueller,..., Scott Bateman, Ian Smith, et al. Grand Challenges in WaterHCI. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2024). Honolulu, USA, Germany. 2024. Association for Computing Machinery, New York, NY, USA, Pages 1 – 18. https://doi.org/10.1145/3613904.3642052 |
| DIS 2024 | Setareh Aghel Manesh, Tianyi Zhang, Yuki Onishi, Kotaro Hara, Scott Bateman, Jiannan Li, and Anthony Tang. How People Prompt Generative AI to Create Interactive VR Scenes. In Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24). 2319–2340. https://doi.org/10.1145/3643834.3661547 |
| BioRob 2024 | Ethan Eddy, Evan Campbell, Scott Bateman and Erik Scheme. Human-Machine Interaction Using Discrete Myoelectric Control: Contrastive Learning Reduces False Activations During Activities of Daily Living. In the Proceedings of IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob). Heidelberg, Germany. September 2024. 6 pages. https://doi.org/10.1109/BioRob60516.2024.10719934 |
| IEEE GEM 2024 | Josh Roberts, Jason Wuertz, Max V. Birk, Scott Bateman and Daniel J. Rea. How the Emotional Content of Music Affects Player Behaviour and Experience in Video Games. In the Proceedings of 2024 IEEE Gaming, Entertainment, and Media Conference (GEM). Turin, Italy, 2024, 6 pages. https://doi.org/10.1109/GEM61861.2024.10585638 |
| IEEE SMC 2023 | Ethan Eddy, Evan Campbell, Scott Bateman and Erik Scheme. Leveraging Task-Specific Context to Improve Unsupervised Adaptation for Myoelectric Control. In the Proceedings of the 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Honolulu, Oahu, HI, USA, 2023, pp. 4661-4666, doi: https://doi.org/10.1109/SMC53992.2023.10394393 |
| CHI 2023 | Ethan Eddy, Erik Scheme, Scott Bateman. A Framework and Call to Action for the Future Development of EMG-Based Input in HCI. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2023). Hamburg, Germany. 2023. 1-23. DOI: 10.1145/3544548.3580962 |
| GI 2023 | Connor Wilson, Daniel J. Rea, Scott Bateman. Temperature Illusions in Mixed Reality using Color and Dynamic Graphics. In the Proceedings of Graphics Interface (GI 2023). Victoria, BC. https://openreview.net/forum?id=U5zU7yDUhKx |

| | |
|---|---|
| CHI 2021 | Aaron Tabor, Scott Bateman, Erik Scheme, Book Sadprasid, m.c. schraefel. Designing Peripheral Breathing Guidance. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2021). Yokohoma, Japan. 2021. 1-13. DOI: 10.1145/3 411 764.3445388. |
| EMBC 2020A | Ian Smith, S. Gill, Scott Bateman, Erik Scheme. "Comparison of Feedback Approaches to Improve Training in Partial Weight-Bearing", *42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Montreal, Canada, July 20-24, 2020. DOI: 10.1109/EMBC44 109.2020.9176207 |
| EMBC 2020B | Jason Chang, Angkoon Phinyomark, Scott Bateman, Erik Scheme. "Wearable EMG-Based Gesture Recognition Systems During Activities of Daily Living: An Exploratory Study", *42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Montreal, Canada, July 20-24, 2020. DOI: 10.1109/EMBC44 109.2020.9176615 |
| CHI PLAY 2020 | Cassandra Folkins, Jeff Mundee, Emily Read, Max V. Birk, Scott Bateman. A Serious Game for Promoting Positive Attitudes Towards Nursing Homes Among Youth. In Proceedings of the ACM International Symposium on Human-Computer Interaction in Play. November 2020, Ottawa, Canada, 12 pages. |
| ISMAR 2020 | Martin Feick, Scott Bateman, Anthony Tang, Nicolai Marquardt, Andre Miede. TanGi: Tangible Proxies for Embodied Object Exploration and Manipulation in Virtual Reality. In Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2020. November 2020, Brazil. 14 pages. |
| GI 2020 | Vinod Asokan, Scott Bateman, Anthony Tang. Assistance for Target Selection in Mobile Augmented Reality. In the Proceedings of Graphics Interface 2020. May 2020, Toronto, Canada. 12 pages. |
| CHI 2019 | Jason Wuertz, Max V. Birk, Scott Bateman. Healthy Lies: The Effects of Misrepresenting Player Health Data on Experience, Behavior, and Performance. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2019). May 2019, Montreal, Canada. Paper 319, 12 pages. https://doi.org/10.1145/3 290 605.3300549 |
| CHI 2018 | Jason Wuertz, Sultan A Alharthi, William A Hamilton, Scott Bateman, Carl Gutwin, Anthony Tang, Zachary Toups, Jessica Hammer. A Design Framework for Awareness Cues in Distributed Multiplayer Games. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2018). April 2018, Montreal, Canada. 243-257. https://doi.org/10.1145/3 173 574.3173817 |
| CHI 2017A | Aaron Tabor, Scott Bateman, Erik Scheme, David Flatla, Kathrin Gerling. Designing Game-Based Myoelectric Prosthesis Training. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2017). May 2017, Denver, USA. 1352-1363. https://doi.org/10.1145/3 025 453.3025676 |
| CHI 2017B | Jason Wuertz, Scott Bateman, Anthony Tang. Why Players use Pings and Annotations in Dota 2. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2017). May 2017, Denver, USA. 1978-1982. https://doi.org/10.1145/3 025 453.3025967 |

| | |
|---|---|
| CSCW 2017 | Carl Gutwin, Scott Bateman, Gaurav Arora, Ashley Coveney. Looking Away and Catching Up: Dealing with Attentional Disconnection in Synchronous Groupware. In the Proceedings of the ACM Conf. on Computer Supported Cooperative Work and Social Software (CSCW 2017). February 2017, Portland, USA. 2221-2235. https://doi.org/10.1145/2998181.2998226 |
| DIS 2017 | Anthony Tang, Omid Farkoufar, Carman Neustaedter, Scott Bateman. Collaboration in 360° Videochat: Challenges and Opportunities. In the Proceedings of the ACM Conf. on Designing Interactive Systems (DIS 2017). June 10-14, 2017. Edinburgh, UK. 1327-1339. https://doi.org/10.1145/3064663.3064707 |
| MEC 2017 | Aaron Tabor, Scott Bateman, Erik Scheme. Quantifying Muscle Control in Myoelectric Training Games. In the Proceedings of the Myoelectric Control Symposium (MEC 17). August 15-18, 2017. Fredericton, NB, Canada. 4 pages. |
| CHI 2016 | Omid Farkourfar, Kevin Ta, Richard Tang, Scott Bateman, Tony Tang. Stabilized Annotations for Mobile Remote Assistance. In the Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2016). May 2016, San Jose, USA. 1548-1560. http://dx.doi.org/10.1145/2858036.2858171 |
| DIS 2016 | Brennan Jones, Kody Dillman, Richard Tang, Anthony Tang, Ehud Sharlin, Lora Oehlberg, Carman Neustaedter, Scott Bateman. Elevating Communication, Collaboration, and Shared Experiences in Mobile Video through Drones. In the Proceedings of the ACM SIGCHI Conference on Designing Interactive Systems (DIS 2016). June 4-8, 2016, Brisbane, Australia. 1123-1135. http://dx.doi.org/10.1145/2901790.2901847 |
| CHIR 2016 | Scott Bateman, Carl Gutwin. (The Lack of) Privacy Concerns with Sharing Web Activity at Work and the Implications for Collaborative Search. In the Proceedings of the ACM SIGIR Conf. on Human-Information Interaction and Retrieval (CHIIR 2016). March, 2016, Chapel Hill, USA. 43-52. DOI: http://dx.doi.org/10.1145/2854946.2854977 |
| CHI 2015A | Brennan Jones, Anna Witcraft, Scott Bateman, Carman Neustaedter, Anthony Tang. Mechanics of Camera Work in Mobile Video Collaboration. In the Proceeding of the ACM Conf. on Human Factors in Computing Systems (CHI 2015). April 2015, Seoul, Korea. 957-966. |
| CHI 2015B | Richard Tang, Anthony Tang, Scott Bateman, Xing-Dong Yang, Joaquim Jorge. Physio@Home: Exploring visual guidance and feedback techniques for physiotherapy exercises. In the Proceeding of the ACM Conf. on Human Factors in Computing Systems (CHI 2015). April 2015, Seoul, Korea. 4123-4132. |
| EICS 2015 | Aditya Nittala, Xing-Dong Yang, Scott Bateman, Ehud Sharlin and Saul Greenberg. PhoneEar: Interactions for Mobile Devices that Hear High-Frequency Sound-Encoded Data. In the Proceedings of the 7th ACM SIGCHI Symposium on Engineering Interactive Computing Systems - EICS'15. (Duisburg, Germany), ACM Press, 6 pages, June 23-26. |
| CHI PLAY 2014 | Jared Cechanowicz, Carl Gutwin, Scott Bateman, Regan Mandryk, Ian Stavness. Improving Player Balancing in Racing Games. In the Proceedings of The ACM SIGCHI Annual |

| | |
|---|---|
| | Symposium on Computer-Human Interaction in Play (CHI PLAY), October 19th-22nd, 2014. Toronto, Ontario, Canada. 47-56. |
| CHI 2014 | Rodrigo Vicencio-Moreira, Regan Mandryk, Carl Gutwin, Scott Bateman. The Effectiveness (or Lack Thereof) of Aim-Assist Techniques in First-Person Shooter Games. Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2014). April 26th-May 1st, 2014. Toronto, ON, Canada. 937-946. |
| CSCW 2013 | Scott Bateman, Carl Gutwin, Gordon McCalla. Supporting Loosely-Coupled Information Seeking in Tightly-Knit Groups using WebWear. Proceedings of the ACM Conf. on Computer Supported Cooperative Work and Social Software (CSCW 2013). Feb. 23rd-27th, 2013. San Antonio, TX, USA. 955-966. |
| AIED 2013 | Graham Erickson, Stephanie Frost, Scott Bateman, Gordon McCalla. Using the Ecological Approach to Create Simulations of Learning Environments. Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2013). July 9-13th, 2013. pp 411-420. |
| CHI 2012 | Scott Bateman, Jaime Teevan, Ryen White. The Search Dashboard: How Reflection and Comparison Impact Search Behavior. Proceedings of the ACM Conf. on Human Factors in Computing Systems (CHI 2012). May 1-5th, 2012. Austin, TX, USA. pp. 1785-1794. |
| UIST 2011 | David Flatla, Carl Gutwin, Lennart Nacke, Scott Bateman, Regan Mandryk. Calibration Games: Making Calibration Tasks Enjoyable by Adding Motivating Game Elements. Proceedings of ACM Conference on User Interface Software Technology (UIST 2011), Santa Barbara, California. October 16-19, 2011. pp. 403-412. |
| CHI 2011 | Scott Bateman, Tadeusz Stach, Regan L. Mandryk, Carl Gutwin. Using Target Assistance for Subtly Balancing Competitive Play. Proceedings of ACM Conf. on Human Factors in Computing Systems (CHI 2011). May 7-12th, 2011. Vancouver, Canada. pp. 2355-2364. **Honorable Mention for Paper Award (top 5% of submissions).** |
| GI 2011 | Scott Bateman, Andre Doucette, Robert Xiao, Carl Gutwin, Regan Mandryk, and Andy Cockburn. Effects of View, Input Device, and Track Width on Video Game Driving. Proceedings of Graphics Interface, May 15-18, 2011. St. John's, Canada. pp. 207-14 |
| CHI 2010 | Scott Bateman, Regan L. Mandryk, Carl Gutwin, Aaron Genest, David McDine, Christopher Brooks. Useful Junk? The Effects of Visual Embellishments on the Comprehension and Memorability of Charts. Proceedings of ACM Conference on Human Factors in Computing Systems (CHI 2010). April 10-15, 2010. Atlanta, GA, USA. pp. 2573-2582. **Best Paper Award (top 1% of submissions).** |
| Group 2009 | Group 2009 Scott Bateman, Michael Muller, Jill Freyne. Personalized Retrieval in Social Bookmarking. Proceedings of the ACM Conference on Supporting Group Work (Group 2009). May 10 - 13, 2009. Sanibel Island, Florida. pp. 91-94. |

| | |
|---|---|
| EICS 2009 | Scott Bateman, Carl Gutwin, Nathaniel Osgood, Gordon McCalla. Interactive Usability Instrumentation. Proceedings of the ACM Symposium on Engineering Interactive Computing Systems (EICS 2009). July 14-17, 2009. Pittsburgh, PA. pp. 45-54. |
| HT 2008 | Scott Bateman, Carl Gutwin, Miguel Nacenta. Seeing Things in the Clouds: The Effect of Visual Features on Tag Cloud Selections. Proceedings of ACM Conference on Hypertext and Hypermedia (HT 2008). Pittsburgh, US. June 16-20, 2008. pp. 193-202. |
| ICSC 2008 | Carlo Torniai, Jelena Jovanovic, Dragan Gasevic, Scott Bateman, Marek Hatala. Leveraging Folksonomies for Ontology Evolution in E-learning Environments. Proceedings of IEEE International Conference on Semantic Computing (ICSC 2008). Santa Clara, CA, USA. August 4-7, 2008. pp. 206-213. |
| ICALT 2008 | Carlo Torniai, Jelena Jovanovic, Dragan Gasevic, Scott Bateman, Marek Hatala. E-Learning Meets the Social Semantic Web. Proceedings of The 8th IEEE International Conference on Advanced Learning Technologies (ICALT 2008). Santander, Cantabria, Spain. July 1-5, 2008. pp. 389-393. |
| ITS 2008 | Jelena Jovanovic, Carlo Torniai, Dragan Gasevic, Scott Bateman, Marek Hatala. Leveraging the Social Semantic Web in Intelligent Tutoring Systems. Proceedings of the Conference on Intelligent Tutoring Systems (ITS 2008). Montreal, Canada. June 23-27, 2008. pp. 563-572 |
| WWW 2007 | Scott Bateman, Christopher Brooks, Gordon McCalla, Peter Brusilovsky. Applying Collaborative Tagging to E-Learning. Workshop Proc. of Tagging and Metadata for Social Information Organization in conjunction with the International World Wide Web Conference (WWW 2007), May 8, 2007. Banff, Canada. 7 pages. |
| AH 2006 | Scott Bateman, Christopher Brooks, Gordon McCalla. Collaborative Tagging Approaches for Ontological Metadata in Adaptive E-Learning Systems. Workshop Proceedings of the 4th International Conference on Adaptive Hypermedia and Adaptive Web-Based Systems (AH 2006), June 20, 2006. Dublin, Ireland. pp. 3-12. |
| LORN 2006A | Scott Bateman, Rosta Farzan, Peter Brusilovsky, Gordon McCalla. OATS: The Open Annotation and Tagging System. Proceedings of the Third Annual International Scientific Conference of the Learning Object Repository Research Network (LORNET 2006). Montreal, November 8-10, 2006. 10 pages. |
| ITS 2006 | Christopher Brooks, Scott Bateman, Gordon McCalla, Jim Greer. Applying the Agent Metaphor to Learning Content Management Systems and Learning Object Repositories. Proceedings of the 8th International Conference on Intelligent Tutoring Systems (ITS 2006). June 26-30, 2006. Jhongli, Taiwan. pp. 808-810. |
| LORN 2006B | Christopher Brooks, Scott Bateman, Wengang Liu, Gordon McCalla, Jim Greer, Dragan Gasevic, Timmy Eap, Griff Richards, Khaled Hammouda, Shady Shehata, Mohamed Kamel, Fakhri Karray, Jelena Jovanovic. Issues and Directions with Educational Metadata. Proceedings of the Third Annual International Scientific Conference of the Learning Object Repository Research Network (LORNET 2006), Montreal, PQ, November 8-10, 2006. 9 pages. |

### 5.3 a) BOOKS

N/A

### 5.3 b) PART(S) OF BOOKS

| | |
|---|---|
| SWEL 2009 | Christopher Brooks, Scott Bateman, Jim Greer, Gord McCalla. Lessons Learned using Social and Semantic Web Technologies for E-Learning. Book Chapter 14 in Semantic Web Technologies for e-Learning, Vol. 4 (The Future of Learning). ISBN: 978-1-60 750-062-9, November 2009. pp. 260 - 278. |

### 5.4 EDITORIAL RESPONSIBILITIES

| | |
|---|---|
| 2025-26 | Foundations of Digital Games (FGD 2026) – Track Chair for the Games Beyond Entertainment Track. Recruited reviewers and provided meta-reviews for 10 papers.<br><br>Graphics Interface (GI 2026) – Committee Member. Reviewed papers as a member of the program committee. |
| 2024-25 | Subcommittee Chair (Blending Interactions Subcommittee) -- ACM Conference on Human Factors in Computing Systems (CHI 2025). – Led the paper editorial process, of ~220 paper submissions, including the organization of associate chairs, reviewers, the review process, program committee meeting, and guiding all acceptance decisions and processes for the subcommittee. |
| 2023-24 | Subcommittee Chair (Blending Interactions Subcommittee) -- ACM Conference on Human Factors in Computing Systems (CHI 2024). – Led the paper editorial process, of ~220 paper submissions, including the organization of associate chairs, reviewers, the review process, program committee meeting, and guiding all acceptance decisions and processes for the subcommittee. |
| 2022-23 | Papers Co-Chair - ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2023) – Paper Chair – led the full paper editorial process, of ~140 paper submissions, including the organization of associate chairs, reviewers, the review process, program committee meetings, and guiding all acceptance decisions and processes. |
| 2021-22 | Papers Co-Chair - ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2022) – Paper Chair – led the full paper editorial process, of ~125 paper submissions, including the organization of associate chairs, reviewers, the review process, program committee meetings, and guiding all acceptance decisions and processes. |
| 2021 | ACM Conference on Human Factors in Computing Systems (CHI 2021) – Associate Chair – responsible for soliciting reviewers for each of 18 research papers, guiding the review process, writing meta-reviews, providing a recommendation and attending the virtual PC meeting. December 6-7, 2020. |

| | |
|---|---|
| 2019 | ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2019) – Associate Chair – solicited reviewers, provided meta-reviews and made acceptance decisions on 9 papers<br><br>ACM Conference on Designing Interactive Systems (DIS 2019) – Associate Chair - responsible for soliciting reviewers for each of 10 research papers, guiding the review process, writing meta-reviews, providing a recommendation and attending the virtual PC meeting.<br><br>ACM Conference on Human Factors in Computing Systems (CHI 2019) – Associate Chair – responsible for soliciting reviewers for each of 17 research papers, guiding the review process, writing meta-reviews, providing a recommendation and attending the virtual PC meeting. December 6-7, 2018. |
| 2014 - 2019 | Member of the Editorial Board (Associate Editor) of the International Journal of Human-Computer Studies (IJHCS). Solicit reviewers, provide (meta-) reviews, make acceptance recommendations, and participate in board meetings (5 years). |
| 2018 | ACM Conference on Human Factors in Computing Systems (CHI 2018) – Associate Chair – responsible for soliciting reviewers for each of 20 research papers, guiding the review process, writing meta-reviews, providing a recommendation and attending the PC meeting in Montreal, PQ.<br>ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2018) – Associate Chair – solicited reviewers, provided meta-reviews, and made acceptance decisions on 8 papers. |
| 2017 | ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2017) – Associate Chair – solicited reviewers, provided meta-reviews, and made acceptance decisions on 10 papers. |
| 2016 | ACM Conference on Human Factors in Computing Systems (CHI 2016) – Associate Chair – responsible for soliciting reviewers for each of 24 research papers, guiding the review process, writing meta-reviews, providing a recommendation, and attending the PC meeting in San Jose California.<br>Graphics Interface (GI 2016) – Associate Chair. Find reviewers and provide reviews & meta-reviews for 5 papers, participate in virtual PC meeting to make acceptance decisions.<br>ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2016) – Associate Chair – solicited reviewers, provided meta-reviews, and made acceptance decisions on 10 papers. |
| 2015 | ACM Conference on Human Factors in Computing Systems (CHI 2015) – Associate Chair – responsible for soliciting three reviewers for each of 11 research papers, guiding the review process, writing meta-reviews, providing a recommendation, and attending the PC meeting in Seoul, Korea.<br>Graphics Interface (GI 2015) – Associate Chair. Find reviewers and provide reviews & meta-reviews for 7 papers, participate in virtual PC meeting to make acceptance decisions.<br>ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2015) – Associate Chair – solicited reviewers, provided meta-reviews and made acceptance decisions on 11 papers. |

| | |
|---|---|
| 2014 | ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2014) – Associate Chair – solicited reviewers, provided meta-reviews and made acceptance decisions on 7 papers.<br><br>Graphics Interface (GI 2014) – Associate Chair. Find reviewers and provide reviews & meta-reviews for 7 papers, participate in PC meeting to make acceptance decisions.<br><br>ACM Conference on Human Factors in Computing Systems (CHI 2014) - Work-In Progress Associate Chair – solicited reviewers, provided meta-reviews and made acceptance decisions for 6 short papers. |
| 2009 | ACM Conference on Human Factors in Computing Systems (CHI 2009) - Work-In Progress Associate Chair – solicited reviewers, provided meta-reviews and made acceptance decisions for 10 short papers. |

### 5.5 CREATIVE ACCOMPLISHMENTS

N/A

### 5.6 NON-REFEREED (OR LIGHTLY REVIEWED) PUBLICATIONS

| | |
|---|---|
| CHI EA 2026 | Ethan Eddy, Erik J Scheme, Géry Casiez, and Scott Bateman. How Low Can Your Thumb Go? Investigating the Precision of Unidirectional Thumb Movements. In Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26). Article 376, 1–6. https://doi.org/10.1145/3772363.3799282 |
| ARCASN 2024 | Renée Gordon, Donald Leidl, Xun Ge, Scott Bateman. 2024. Evaluating an Immersive Virtual Reality Simulation Intervention for Distributed Delivery of a Nursing Program. Atlantic Region Canadian Association of Schools of Nursing (ARCASN) Conference, Ways of Kowing: Building Partnerships with Indigenous and Nursing Knowledge to Educate Future Nurses, University of Prince Edward Island School of Nursing, Charlottetown, PEI. |
| CASN 2024 | Renée Gordon, Donald Leidl, Xun Ge, Scott Bateman. 2024. The Evolution of Immersive Virtual Reality Simulation for Distributed Delivery of an Undergraduate Nursing Program. Canadian Association of Schools of Nursing (CASN) Biennial Canadian Nursing Education Conference. Calgary, AB. |
| CHI PLAY EA 2022 | Book Sadprasid, Aaron Tabor, Erik Scheme, Scott Bateman. 2022. Focus Cat: Designing Idle Games to Promote Intermittent Practice and On-Going Adherence of Breathing Exercise for ADHD. In Extended Abstracts of the 2022 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '22). 303–310. DOI: 10.1145/3505270.3558381<br><br>Lieke Vermeulen, Max V. Birk, Scott Bateman, Henk S. Schipper, Sanne L. Nijhof, and Yuan Lu. 2022. "It's All in the Game" A board game to facilitate disease-related conversations between children with a chronic disease and their peers. In Extended Abstracts of the 2022 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '22). Association for Computing Machinery, New York, NY, USA, 140–145. 10.1145/3505270.3558314 |

| | |
|---|---|
| CHI EA 2022 | Christal Clashing, Maria Fernanda Montoya Vega, Ian Smith, Joe Marshall, Leif Oppermann, Paul H Dietz, Mark Blythe, Scott Bateman, Sarah Jane Pell, Swamy Ananthanarayan, and Florian Floyd Mueller. 2022. Splash! Identifying the Grand Challenges for WaterHCI. In Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 78, 1–6. https://doi.org/10.1145/3 491 101.3503723 |
| CHI PLAY EA 2021 | Book Sadprasid, Ethan Eddy, Aaron Tabor, Erik Scheme, and Scott Bateman. 2021. Percussion Hero: A Chest Physical Therapy Game for People with Cystic Fibrosis and their Caregivers. In Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '21). Association for Computing Machinery, New York, NY, USA, 361–367. https://doi-org.proxy.hil.unb.ca/10.1145/3 450 337.3483508<br>Best Game in the Student Game Design Competition. |
| CHI EA 2021 | m.c. schraefel, Josh Andrés, Aaron Tabor, Scott Bateman, Abby Wanyu Liu, Mike Jones, Kai Kunze, Elizabeth Murnane, and Steeven Villa. 2021. Body As Starting Point 4: Inbodied Interaction Design for Health Ownership. In Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 85, 1–5. DOI: https://doi.org/10.1145/3 411 763.3441335 |
| CHI EA 2020 | Aaron Tabor, Ian C. J. Smith, Scott Bateman, Josh Andres, Andrés Mejía Figueroa, and m.c. schraefel. 2020. 3rd Body As Starting Point Workshop: Exploring Themes for Inbodied Interaction Research and Design. In Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA 20). Association for Computing Machinery, New York, NY, USA, 1–8. https://doi.org/10.1145/3 334 480.3375170 |
| CHI PLAY EA 2020 | Aaron Tabor, Reyhan Pradantyo, Book Sadprasid, Max V. Birk, Erik Scheme, and Scott Bateman. 2020. Bubble Breather - A Breathing Exercise Game to Support Pneumonia Rehabilitation and Recovery. In Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '20). Association for Computing Machinery, New York, NY, USA, 86–90. https://doi.org/10.1145/3 383 668.3419921 Best Interactive Experience in the Student Game Design Competition. |
| Ubicomp 2019A | Connor Wilson, Aaron Tabor, Scott Bateman. Diversifying Pro-Environmental Behaviours Redesigning Applications to Incorporate Environmental "Spill-over". In the Proceedings of the Workshops of Ubicomp 2019. London, UK. Sept. 9, 2019. 4 pages. |
| Ubicomp 2019B | Aaron Tabor, Connor Wilson, Scott Bateman. Designing to Support Uncomfortable Breathing Exercises: Ethical Considerations. In the Proceedings of the Workshops of Ubicomp 2019. London, UK. Sept. 9, 2019. 4 pages. |
| BigVis 2019 | Alex Watson, Scott Bateman, Suprio Ray. PySnippet: Accelerating Exploratory Data Analysis in Jupyter Notebook through Facilitated Access to Example Code. In the Proceedings of the Workshops of the EDBT/ICDT 2019 Joint Conference (EDBT/ICDT 2019). Lisbon, Portugal, March 26, 2019. 4 pages. http://ceur-ws.org/Vol-2322/BigVis_8.pdf |

| | |
|---|---|
| UIST 2018 | Martin Feick, Anthony Tang, and Scott Bateman. 2018. Mixed-Reality for Object-Focused Remote Collaboration. In The 31st Annual ACM Symposium on User Interface Software and Technology Adjunct Proceedings (UIST '18 Adjunct). ACM, New York, NY, USA, 63-65. DOI: https://doi.org/10.1145/3 266 037.3266102 |
| Tech Report 2018 | Scott Bateman, Carl Gutwin, Manasi Shah, Nathaniel Brewer. Supporting Visual Search and Spatial Memory in a Mobile Application Launcher. University of New Brunswick Technical Report. TR18-240. October 2018. |
| Sci. Atlantic 2017 | Manasi Shah, Nathaniel Brewer, Scott Bateman, Carl Gutwin. SpaceLaunch: A Faster App Launcher that Leverages Spatial Memory. Abstract presented at the Science Atlantic Computer Science Conference. Fredericton, NB. Oct. 13th-14th, 2017. My student Shah was awarded the Science Communication Award. |
| Sci. Atlantic 2017 | Nathaniel Brewer, Scott Bateman. The Development of SpaceLaunch: A Faster App Launcher Developed in Unity. Abstract presented at the Science Atlantic Computer Science Conference. Fredericton, NB. Oct. 13th-14th, 2017. My student Brewer was awarded 2nd best oral presentation. |
| Sci. Atlantic 2017 | Jason Wuertz, Scott Bateman. Understanding the Design of Non-Verbal Communication Tools through Studying Games. Abstract presented at the Science Atlantic Computer Science Conference. Fredericton, NB. Oct. 13th-14th, 2017. |
| Sci. Atlantic 2017 | Jawad Refai, Scott Bateman, Michael Fleming. Assistance Techniques in Video Games. Abstract presented at the Science Atlantic Computer Science Conference. Fredericton, NB. Oct. 13th-14th, 2017 |
| Sci. Atlantic 2017 | Ian Smith, Santinder Gill, Erik Scheme, Scott Bateman. Improving gait through sensored cane feedback. Abstract presented at the Science Atlantic Computer Science Conference. Fredericton, NB. Oct. 13th-14th, 2017. |
| CHI Play WIP 2016 | Aaron Tabor, Scott Bateman, Erik Scheme. Game-Based Myoelectric Training. In the Extended Abstracts of the ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2016). October 16-19, 2016. Austin, TX. 299-306. https://doi.org/10.1145/2 968 120.2987731 |
| Sci. Atlantic 2015 | Jason Wuertz, Nikita Volodin, Scott Bateman. DOTA 2 as Platform for Human-Computer Interaction Research. Abstract presented at the Science Atlantic Computer Science Conference. Wolfville, NS. Oct. 23rd-25th, 2015. My student (Wuertz) was awarded 2nd best oral presentation. |
| Sci. Atlantic 2014 | Nikita Volodin, Scott Bateman, Carl Gutwin. A Flexible Programming Framework for Multi-Display Environments. Abstract presented at the Science Atlantic Computer Science Conference. Saint John, NB. Oct. 3rd-5th, 2014. My student Volodin was awarded 2nd best oral presentation. |

| | |
|---|---|
| CHI-WIP 2014 | Richard Tang, Hesam Alizadeh, Anthony Tang, Scott Bateman, Joaquim Jorge. Physio@ Home: design explorations to support movement guidance. In the Extended Abstracts on Human Factors in Computing Systems (CHI'14), 1651-1656. |
| CSCW 2012 | Expressiveness Matters in Command & Control Visualization. Proceedings of the Collaboration and Crisis Informatics Workshop, held at the 2012 ACM Conference on Computer Supported Collaborative Work (CSCW 2012). February 11-15th, 2012, Seattle, WA., USA. 4 pages. |
| HCIR 2011 | Scott Bateman, Carl Gutwin, Gordon McCalla, Ryen White. Collective Information Seeking: Supporting Search with Low-Cost Activity Sharing and Collaboration. Proceedings of the Fifth Workshop on Human-Computer Interaction and Information Retrieval (HCIR 2011). Oct. 2011. Mountain View, CA, USA. 4 pages. |
| CSCW DC 2010 | Scott Bateman. (2010) Using Group Interaction History in the Wild. Extended Proceedings of the Doctoral Colloquium of the 2010 ACM Conference on Computer Supported Collaborative Work (CSCW 2010). February 6 –10, 2010. Savannah, USA. pp. 523 - 524. |
| HCI-ED 2010 | Christopher Brooks, Scott Bateman, Jim Greer, Carl Gutwin. Improving the Video Timeline for Educational Video Content. In the Proceedings of the Workshop on HCI and Education, held at the ACM Conference on Human Factors in Computing Systems (CHI 2010), April 10-15, 2010. Atlanta, GA, USA. 4 pages |
| Tech Rep. 2009 | Scott Bateman, Regan Mandryk, Carl Gutwin, Robert Xiao. Investigation of Targeting-Assistance Techniques for Distant Pointing with Relative Ray Casting. Technical Report 3-2009, Department of Computer Science, University of Saskatchewan. 8 pages. Published December 14, 2009. |
| ITS-W 2008 | Scott Bateman, Christopher Brooks, Gord McCalla, Jim Greer. Learning Object Metadata: From the locally prescribed to the socially derived. In the Workshop for Semantic-Web Resource Repository Design, Search, and Management. Conference on Intelligent Tutoring Systems (ITS 2008). June 23, 2008. Montreal, Canada. 10 pp. |
| SWUI 2008 | Scott Bateman, Jelena Jovanovic, Carlo Torniai, Dragan Gasevic, Marek Hatala. Combined Usage of Ontologies and Folksonomies in E-learning Environments. In the Proceedings of the Semantic Web User Interaction Workshop held in conjunction with ACM Conference on Human Factors in Computing Systems (CHI 2008). Florence, Italy. April 5, 2008. 4 pages. |
| LORNET 2006C | Scott Bateman, and Gordon McCalla Collective Intelligence in E-Learning. Technical System Demonstration in the Demo Competition at the International Scientific Conference of the Learning Object Repository Research Network (LORNET 2006). Montreal, Canada. November 8-10, 2006. Awarded 1st Place in the Demo Competition. |

**OTHER SUPERVISED STUDENT CONTRIBUTIONS**

| | |
|---|---|
| CHI Play SDC 2016 | Aaron Tabor, Alex Kienzle, Carly Smith, Alex Watson, Jason Wuertz, David Hanna. The Falling of Momo: A Myo-Electric Controlled Game to Support Research in Prosthesis Training. Paper and demonstration accepted in the Student Design Competition Track at ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2016). October 16-19, 2016. Austin, TX. ACM Press. ***Awarded Best Game in the Student Demo Competition.*** |

### 5.7 a) MANUSCRIPTS SUBMITTED

N/A

### 5.7 b) MANUSCRIPTS IN PREPARATION

Five manuscripts in preparation for various journal or conference submissions.

### 5.8 PAPERS PRESENTED AT PROFESSIONAL AND TECHNICAL MEETINGS

Note: All conference publications listed above result in an invited presentation at the conference. In all cases first authors of the above conference papers presented at the conference.

### 5.9 PRESENTATION OF SEMINARS, CLINICS, WORKSHOPS

| | |
|---|---|
| 2026 | AI for All in Atlantic Canada: Creating Conditions for Thriving in a Peripheral Region. Invited talk presented at the Eindhoven AI System Institute (EAISI), Technical University Eindhoven (TU/E), Eindhoven, Netherlands. June 22, 2026. |
| 2025 | Human Oversight of AI is Necessary and Fast(-ish): Ensuring Responsible AI in Public Services. Invited talk at the Open Data and GovMaker Summit, Fredericton, NB. November 26, 2025. |
| 2024 | Shaping Human Performance for a Better (HCI) Future: Explorations in Games, Visualization, Coding & Mixed Reality. Invited Talk presented at the University of Waterloo. September 19th, 2024. |
| 2023 | Shaping Human Performance for a Better (HCI) Future: Explorations in Games, Visualization, Coding & Mixed Reality. Invited Talk presented at Singapore Management University. December 13, 2023.<br>Developing and Evaluating Mixed Reality HCI Systems. Three-day workshop developed and presented at the Singapore Management University December 17-19, 2023. |
| 2022 | HCI Research at the University of New Brunswick: Mixed Reality, Robots, Games, and Health. Invited talk presented at the Hochschule Bonn-Rein-Sieg. St. Augustin, Germany. December 13, 2022. |

Research in the UNB HCI Lab: The First 7 years. Invited talk presented to the Loki Research group at Inria, Lille, France. July 4, 2022.

2018 Can video games inform the design of rehabilitation technology? Invited talk in the Institute of Biomedical Engineering seminar series. UNB, Fredericton, NB. November 14, 2018.

ENGspire. Public panel presentation on path to a research career. Invited by the Atlantic Student Research Journal. Fredericton, NB. October 20, 2018.

Sketching for System Design. Public tutorial presented at the NextGenUX Workshop and cast live online. Fredericton, NB. April 19-20, 2018.

2017 Research in the UNB HCI Lab. Public Seminar presented to the School of Mathematical and Computational Sciences at the University of Prince Edward Island, Nov. 27, 2017.

Serious-ification: Doing "Serious" HCI Research… Based on Video Games. Invited talk for the Dept. of Computer Science at the University of Calgary. May 2017.

2015 An NSERC Engage Success Story: A collaboration between UPEI and nGauge, Inc. Presentation to the NSERC Atlantic Advisory Committee. May 2015, Charlottetown, PEI.

Cloud Computing. Presentation and panel discussion for the Charlottetown Chapter of Canada's IT Professional Society (CIPS) meeting April 2015.

2014 Why PEI for IT? Presentation and panel discussion for the Charlottetown Chapter of Canada's IT Professional Society (CIPS) meeting April 2014.

2013 Social Feedback: Designing Interfaces for Learning from Shared Interaction History. Department of Computer Science Seminar, University of Calgary, Calgary, AB, June 18th, 2013.

The Science of Fun Interactions. Invited talk presented to Spielo International, Moncton, NB, January 25th, 2013.

2012 Social Feedback: Learning from Shared Interaction History. Invited talk presented to the Interactive Information Retrieval Group at University of North Carolina, Chapel Hill, N.C., November 5th, 2012.

Social Feedback: Learning from Shared Interaction History to Support Information Seeking on the Web. Department of Computer Science Seminar, University of Saskatchewan, Saskatoon, SK, August 2nd, 2012.

Social Feedback: Learning from Shared Interaction History. Invited talk presented at Microsoft Research, Redmond, WA, April 5th, 2012.

Target Assistance and Social Feedback: Two Approaches for Improving Performance with Interactive Systems. Invited talk presented at AT&T Research, Florham Park, NJ, March 26th, 2012.

Social Feedback: Learning from Shared Interaction History. Invited talk presented to the Collaborative User Experience Group, IBM Research, Cambridge, MA, January 30th, 2012.

2010 What I Wish I Had Known Before I Started as a Graduate Student Sessional Instructor. Invited Panelist. Presented to the Graduate Student Teaching and Learning Workshop, University of Saskatchewan, September 1, 2010.

2007 Effectively Describing and Using Web Resources Invited talk presented to the PAWS Lab, at the University of Pittsburgh, April 2, 2007.

2007 The Web After 20 Years: A Convergence or Divergence of Ideas. Invited talk presented at the Saskatchewan Interactive Conference, Saskatoon, Saskatchewan, March 8, 2007.

2006 How Can We Describe and Use Digital Learning Materials More Effectively? Invited talk presented to Adaptive Web Systems Group at the University of Leeds, UK, September 5, 2006.

## 5.10 CONSULTING ACTIVITIES, PROFESSIONAL SERVICES TO GOVERNMENTAL, PROFESSIONAL & INDUSTRIAL ASSOCIATIONS, EDUCATION INSTITUTIONS, GRANTING AGENCIES, ETC.

### EXTERNAL AWARD COMMITTEE MEMBER

2025 Selection committee co-chair for the CHCCS/SCDHM Graphics Interface Early Career Researcher Award. January - March 2025.

2024 Selection committee co-chair for the CHCCS/SCDHM Graphics Interface Early Career Researcher Award. January - March 2024.

2022 Selection committee member for the CHCCS/SCDHM Graphics Interface Early Career Researcher Award. January 2022.

2021 Selection committee member for the CHCCS/SCDHM Graphics Interface Early Career Researcher Award. February 2021.

### EXTERNAL PROGRAM REVIEWER

2018 Academic External Reviewer for the Bachelor of Information Technology in Interactive Media and Design Program, Carleton University, November 2018.

### PROMOTION ASSESSMENT

2026 Carleton University. Assessed promotion case for Faculty member applying for promotion to Associate Professor.

2026 Waterloo University. Assessed promotion case for Faculty member applying for promotion to Associate Professor.

2025 Dalhousie University. Assessed promotion case for Faculty member applying for promotion to Associate Professor.

2024 University of Prince Edward Island. Assessed promotion case for a Faculty member applying for promotion to Full Professor.

2020 Carleton University. Assessed promotion case for a Faculty member applying for promotion to Associate Professor.

## EXTERNAL GRANT REFEREEING

2026 NSERC Discovery NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 3 applications.

2026 MITACS Accelerate Grant Review. Reviewed applications for a Canadian faculty member's application to the MITACS Accelerate programme. 3 applications.

2025 SNF (Swiss National Science Foundation) National Centre of Competence in Research (NCCR) application review. Reviewed multi-year, multi-university, multimillion Swiss franc application. Re-review of previous application.

2025 NSERC Discovery NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 3 applications.

2024 SNF (Swiss National Science Foundation) National Centre of Competence in Research (NCCR) application review. Reviewed multi-year, multi-university, multimillion Swiss franc application.

2024 NSERC Discovery NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 3 applications.

2023 NSERC Discovery NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 3 applications.

2022 NSERC Discovery NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 3 applications.

Canadian Foundation for SInnovation. CFI John R. Evan's Leaders Fund. Reviewed an application of a Canadian faculty member.

| | |
|---|---|
| 2021 | NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 3 applications. |
| | NSERC Alliance Grant Review. Reviewed Canadian faculty member's application to NSERC. |
| 2020 | NSERC I2I Grant Review. Reviewed Canadian faculty member's application to NSERC. |
| | MITACS Accelerate Grant Review. Reviewed applications for a Canadian faculty member's application to the MITACS Accelerate programme. 3 applications. |
| 2019 | NSERC Collaborative Research and Development Grant Review. Reviewed applications for a Canadian faculty member's application to the NSERC CRD programme. 1 application. |
| | MITACS Accelerate Grant Review. Reviewed applications for a Canadian faculty member's application to the MITACS Accelerate programme. 2 applications. |
| | NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 2 applications. |
| 2018 | MITACS Accelerate Grant Review. Reviewed applications for a Canadian faculty member's application to the MITACS Accelerate programme. 2 applications. |
| | NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 2 applications. |
| 2017 | NSERC Strategic Partnership. Reviewed application to NSERC's Strategic Partnership programme. 1 application. |
| | NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant programme. 1 application. |
| 2016 | 2016 MITACS Converge Grant Review. Reviewed applications for a Canadian faculty member's application to the MITACS Converge programme. 1 application. |
| | 2016 MITACS Accelerate Grant Review. Reviewed applications for a Canadian faculty member's application to the MITACS Accelerate programme. 1 application. |
| | 2016 NSERC Discovery Grant Review. Reviewed application for a Canadian faculty member's application to the Discovery Grant program. 1 application |
| 2014 | Canadian Foundation for Innovation Grant Review Committee Member. Reviewed 6 applications for major infrastructure grants ($2-$16 million) for several Canadian Universities. Attended and discussed applications with the review committee in Toronto (Sept. 2014). |

### 5.11 AWARDS (GRANTS, CONTRACTS, FELLOWSHIPS)

| | |
|---|---|
| 2026 – 2031 | NSERC Discover Grant. Developing Calm, Focused Human-Computer Interactions for Sustainable Productivity (5 years; $210,000; 100% of award). |

| | |
|---|---|
| 2026 | MITACS Accelerate Internship to support Improving Response Quality in RAG-Based AI Systems for General Data IT Professionals Ltd. ($15,000; 50% of award). |
| 2026 | Canadian Foundation for Innovation. John Evans Leaders Fund. Machine Learning Health Vault. (1 year; $97,026; 66% of award). |
| 2026 | Research NB. CFI Matching Funds. Machine Learning Health Vault. (1 year; $97,026; 66% of award). |
| 2025-2026 | MITACS Business Strategy Internship to support Development of Teacher Support Tools for BlueKit AI. ($60,000; 100% of award). |
| 2025 - 2026 | ACOA funded project for the 3D Maritime Digital Twin. ($670,258; 15% of award).<br>MITACS Business Strategy Internship to support Software and DevOps for ClinicGlide Technologies. ($90,000; 100% of award). |
| 2026 | MITACS Business Strategy Internship to support Vibe Check AI: Agentic Social Media Lead Platform for Angry Butterfly. ($30,000; 50% of award).<br>Research contract for the extension of Vibe Check AI with Angry Butterfly. ($9,000; 100% of contract). |
| 2020 - 2026 | NSERC Discovery Grant. Mixed Reality Groupware. (6 years – 5 years plus 1 year extension; $246,000 total; 100% of award) |
| 2024 - 2026 | NBIF Strategic Opportunities Funds to support the establishment of Spatial Computing Education, Training and Research Laboratory (SPECTRAL). ($99,000; 100% of award) |
| 2025 | MITACS Business Strategy Internship to support Development of New Strategies for L'Arche with L'Arche. ($15,000; 100% of award)<br>MITACS Business Strategy Internship to support Mapping clusters to behavior profiles with IGT. ($15,000; 100% of award) |
| 2023-2025 | SSHRC Partnership Grant to Support Able Village (Game-Based Activities for Wellness of Older Adults) with national partners including Loch Lomond Villa, Saint John, NB. ($198,000; 15% of award). |
| 2024 | NBIF Early-Stage Commercialization Fund for the Mixed Reality IV Task Trainer project ($50,000; 100% of award).<br>MITACS Business Strategy Internship to support GeoMix Reality System for Geographic Information Access in Mixed Reality with Kognitiv Spark. ($15,000; 100% of award) |
| 2023-2024 | ACOA Artificial Intelligence Fund for the establishment of the UNB Spatial Computing Education, Training, and Research Lab (SPECTRAL). (2 years; $1,073,125; 100% of award). |

This award consisted of $823,125 from ACOA, $150,000 from Kognitiv Spark, $100,000 from UNB.

2022 – 2023 MITACS Accelerate to support Mixed Reality to Support QA Inspection of Precast Concrete Forms with Strescon, Inc. (8 months; $30,000; 100% of award; co-applicants Zhen Lei and Lloyd Waugh).

2022 MITACS Business Strategy Internship to support A User-Informed Approach to Archaeological Mapping Software with ArchaeoSoft Inc. ($150,000; 100% of award)

2021 - 2023 NBIF Lab-to-Market for a Digital Field Notes Tool for Archaeology. (2 years; $40,000; 100% of award).

2020 - 2024 NSERC CRD. To support Reducing Cognitive Load and Increasing Usability in Mixed Reality Collaboration with Kognitiv Spark (4 years; $150,000 total; 100% of award).

2022-2023 MITACS Accelerate PDF. The development of a mobile mapping tool for archaeologists in collaboration with ArchaeoSoft Inc. To support post-doctoral fellow, Alec McLellan (8 months; $40,000; 100% of award).

NBIF AI Pre-Voucher an AI Approach to Electronic Gaming Machine (EGM) Game Modeling: Reward Function with IGT Canada Solutions ULC. Supervised research of research assistant Jeff Mundee (1 year; $20,000; 100% of award).

Mental Health Research Commission and New Brunswick Health Research Foundation to support "Acceptance and Commitment Therapy through Tele-Counselling for Caregivers in New Brunswick" Durepos, P., MacBean, C., Cormier, N., Gould, O., Olthuis, J., Bateman, S., McCloskey, R., Boamah, S., Sussman, T., Gardner, P., Robitaille, A., Kaasalainen, S. (1 year; $50,000; 0% of award).

2021 MITACS Accelerate. To support post-doctoral fellow, Cora Woolsey, and her participation in the Lab2Market research commercialization program ($15,000; 100% of award).

2020 - 2021 New Brunswick Health Research Foundation & New Brunswick Innovation Foundation - COVID-19 Research Fund. Designing Breathing Games to Support Remote and Virtual Recovery of COVID-19 (total value: $38,000; 50% of award; with Erik Scheme).

2019 - 2021 MITACS Accelerate. To develop patient-facing information displays for VeroSource, Fredericton, NB (total value: $333,333; 15% of award).

NBIF Concept Innovation Grant. To support research project Improving Community Engagement in the Maker Movement ($55,000; 100% of award).

MITACS Accelerate. To support post-doctoral fellow, Cora Woolsey, to support Community Engagement in the Maker Movement with The Ville Cooperative ($55,000; 100% of award).

2014 - 2020 NSERC Discovery Grant. To investigate the Use of Social Feedback in Information Work ($150,000; $25,000/year; 100% of award).

2018 - 2020 MITACS Accelerate Cluster. To develop interactive gesture detection and use for wrist worn EMG device for Pison, Boston, USA (total value: $240,000; 25% of award).

MITACS Accelerate. To support research intern Cassandra Folkins, to develop an educational serious game for New Brunswick Association of Nursing Homes (total value: $40,000; 50% of award).

2019 Research Contract. To investigate interactive surface identification in heads-up-display technology for Kognitiv Spark ($48,000; 100% of contract).

2018 NSERC Engage. To investigate the use and representation of deixis in remote augmented reality-based collaboration ($25,000; 100% of award).

Technology Innovation Grant. National Research Council Canada, for investigation of Drone technology for Flaight, Inc (Fredericton, NB) ($5,250; 100% of contract).

2015 - 2017 NBIF Start-up Award. Start-up funding to support infrastructure and students at the University of New Brunswick ($100,000; 100% of award).

2016 NBIF Innovation Voucher. New Brunswick Foundation for Innovation research contract to investigate information visualization technology with Remsoft, Inc (Fredericton, NB) ($27,758; 100% of award).

Technology Innovation Grant. National Research Council Canada, for investigation information visualization technology with EvaluPlan Consulting (Fredericton, NB) ($5,250; 100% of contract).

2014-2015 Tri-Council New Award Recognition. Awarded by UPEI in support of continued research success (1 course release/year; 5 years). Declined after year 1.

NSERC SurfNet Special Funding. For the creation and support of the Web Application Framework for Multi-Surface Games (GAMS API), with Carl Gutwin (Univ. of Saskatchewan) ($25,000; 50% of award).

NSERC Engage Grant. For investigating the use of Crowdsourcing to identify sales leads with nGauge, Inc. ($24,604.00; 100% of award)

2015 Canadian Excellence Research Chair (CERC) Seed Grants. For Model Support for Treatment Strategies in Managing Sea Louse Parasites and Evolution of Resistance on Atlantic Salmon Farms. Revie, C. (PI), Co-investigators: MacEwan, G., Groner, G., Fast, M., and Bateman, S. ($19,948; 0% of award)

NSERC Engage Grant. For investigating augmented reality in mobile video collaboration with XMG Studio, Inc. (Toronto, ON) ($24,875.00; 100% of award)

Technology Innovation Grant. National Research Council Canada, for investigation object recognition through web technology with XMG Studio, Inc. (Toronto, ON) ($5,250; 100% of contract).

| | |
|---|---|
| 2013 - 2014 | NSERC SurfNet Special Funding. For the creation and support of the Web Application Framework for Multi-Surface Environments (WAMS API), with Carl Gutwin (Univ. of Saskatchewan) ($25,000; 50% of award). |
| 2014 | GRAND Grant. To support projects within the GRAND National Centre of Excellence. ($23,669.00; 100% of award). |
| | Technology Innovation Grant. National Research Council Canada, for investigation of the design of a crowd-funding equity platform with industrial partner Rocklyn Foundation. (Moncton, NB) ($5,250; 100% of contract). |
| | Technology Innovation Grant. National Research Council Canada, for investigation of Energy Consumption Feedback Appliance with industrial partner Reinvented, Inc. (Charlottetown, PE) ($5,250; 100% of contract). |
| 2013 | Young Network Investigator. GRAND National Centre of Excellence ($5,000; 100% of award). |
| | Technology Innovation Grant. National Research Council Canada, for investigation of mobile interfaces with industrial partner Hotpot Parking (Fredericton, NB) ($2,625; 50% of contract) with David Leblanc (UPEI). |
| | Technology Innovation Grant. National Research Council Canada, for investigation of mobile interfaces with industrial partner nGauge (Moncton, NB) ($3,500; 100% of contract). |
| | Major Research Grant. University of Prince Edward Island. Social Feedback to Support Computer Programming ($6,600; 100% of award). |
| | Collaborating Researcher. SurfNet NSERC Strategic Network. |
| 2012 | Start-up Grant. University of Prince Edward Island ($9,500; 100% of award). |

### 5.12 OTHER EVIDENCE OF RESEARCH ACCOMPLISHMENTS

N/A

### 5.13 MEMBERSHIP AND ACTIVE INVOLVEMENT IN PROFESSIONAL AND LEARNED SOCIETIES

Member of the ACM – Association of Computer Machinery

Member of ACM SIGCHI – Special Interest Group of Human-Computer Interaction

### 5.14 RECORD OF PURSUIT OF ADVANCED DEGREES AND/OR FURTHER ACADEMIC STUDY

N/A

## SECTION 6 | **SERVICE**

### 6.1 SERVICE TO THE UNIVERSITY

- **AI in Research Expert Panel –** participated in a University panel on advising faculty, staff, and students on the use of Generative AI in all aspects of research work. 2025.
- **Equity, Diversity, and Inclusion Training Offered University Wide** – I mentored a student-led initiative to create and offer EDI training to faculty and staff. The training was attended by faculty, staff and students from across the university. 2021, 2022.

**Committees (list level, committee name, dates)**

UNB, Faculty of CS – Graduate Committee Member (2018-19, 2019-20, 2020-21, 2021-22, 2022-23, 2023-24, 2024-25)

UNB, Faculty of CS – Policy and Planning Member (2023-24, 2024-25)

UNB – Inter-Faculty Committee (IFAC) Member (2017-18)

UNB, Faculty of CS – Vision Committee Member (2017-18, 2018-19)

UNB, UNB, Faculty of CS – Second Year Advisor (2019-20, 2020-21, 2021-22, 2022-23, 2023-24)

UNB, Faculty of CS – CS Square Committee (2019-20)

UNB, Faculty of CS – Assessment Committee Member (2017-18, 2019-20)

UNB, Faculty of CS – Science Atlantic Committee (2015-19)

UNB, Faculty of CS – Programming Team Committee (2015-19)

UNB, Faculty of CS – Recruitment Committee (2015-16, 2016-17)

UNBT – FCS Representative for the Collective Agreement Bargaining Committee (2016)

UNBT – FCS Representative for the Collective Agreement Bargaining Committee (2020)

UNB – Senate Admissions Committee (2015-16)

UNB – Institute of Biomedical Engineering Director Search/Hiring Committee (2019)

UNB – Institute of Biomedical Engineering CARE Center Operations Committee (2019-20, 2020-21, 2021-2022)

**Administrative Positions (list position, dates)**

| | |
|---|---|
| July 2024 – present | **Scientific Director –** Research Institute of Data Science and Artificial Intelligence (RIDSAI) \| University of New Brunswick |
| May 2023 – present | **Director –** SPECTRAL Spatial Computing Research Centre Faculty of Computer Science \| University of New Brunswick |
| August 2022 – December 2022 | **Acting Assistant Dean** (Research and Graduate Studies) Faculty of Computer Science \| University of New Brunswick |

### 6.2 SERVICE OUTSIDE THE UNIVERSITY OF SCHOLARLY OR ACADEMIC SIGNIFICANCE

| | |
|---|---|
| 2023-present | Steering Committee Member for ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play. The steering committee is engaged in defining and shaping CHI PLAY's direction, by determining the overall strategy and vision for the conference series. |

## SECTION 7 | **OTHER RELEVANT INFORMATION**

### 7.1 TECHNICAL CONTRIBUTIONS

The technical developments used in research in own work and the work of my research group the HCI Lab is posted publicly to our Github repositories (software repositories) publicly as open-source projects. This is to promote the dissemination of knowledge and the replicability of our research findings. Please see: https://github.com/scottbateman, https://github.com/hcilab, and https://github.com/UNB-SPECTRAL

### 7.2 OUTREACH

| | |
|---|---|
| 2025 | Augmented Reality Smart Glasses are the Cell Phones of the Future. Invited talk at the Fredericton Rotary Club. May 2025. |
| 2025 | Augmented Reality Smart Glasses are the Cell Phones of the Future. Invited talk at the Probus Fredericton Meeting. April 2025. |
| 2022 | Computer Science Video Game Summer Camp and Maker Camps. Mentored the organization of three, one-week summer camps for middle and high school-aged children. Teaching the fundamentals of computer science through fun and engaging video game and electronics-based activities. |
| 2020 | Computer Science Video Game Summer Camp and Maker Camps. Organized six, one-week summer camps for middle and high school-aged children, in collaboration with the Ville Cooperative. Teaching the fundamentals of computer science through fun and engaging video game and electronics-based activities. Cancelled due to COVID-19. |
| 2019 | Computer Science Video Game Summer Camp. Organized four, one-week summer camps delivered to ~50 middle and high school-aged children. Teaching the fundamentals of computer science through fun and engaging video game-based activities. |
| 2018 | Computer Science Video Game Summer Camp. Organized two, one-week summer camps delivered to 38 middle school-aged children. Teaching the fundamentals of computer science through fun and engaging video game-based activities. |

2017 Interacting with Software. Interactive presentation and demo of interactive software and research activities in the UNB Human-Computer Interactions Lab. Presented to grade 8 class from Woodstock, NB.

2016 Introduction to Computer Science with SonicPi. Prepared and delivered and introductory programming exercise for approximately 60 middle school students at George Street Elementary School, Fredericton, NB.

Introduction to Programming through Data Analysis. Prepared and delivered an introductory programming exercise for two groups of middle school and high school students, attending the Fredericton River Valley Regional Science Fair.

2016 Interacting with Software. Interactive presentation and demo of interactive software and research activities in the UNB Human-Computer Interactions Lab. Presented to grade 8 class from Bliss Carman School.

2015 Hour of Code Mentor. Volunteered to provide hour of code mentorship and instruction to a grade two class (Mrs. MacKay) at New Maryland Elementary School.

Making Games with Scratch. Introductory tutorial and hands-on activities about how to use Scratch to make games. Presented to grades 5 and 6 at Spring Park School, Charlottetown.

Hour of Code Mentor. Volunteered to provide hour of code mentorship and instruction to a grade two class (Mrs. MacKay) at New Maryland Elementary School.

Making Games with Scratch. Introductory tutorial and hands-on activities about how to use Scratch to make games. Presented to grades 5 and 6 at Spring Park School.

2014 The Charlottetown Maker Co-operative. Established a group of interested faculty and members of the community to establish a community-based maker space in Charlottetown. The maker space opened in 2015 and offer community run courses in computer programming, electronics, 3D printing and making.

2013 Introduction to Programming Games in Scratch and Python. Outreach course. I designed and taught an outreach course for fifteen children, aged 12-15. The course taught introductory programming skills and exposed students to Computer Science. The course consisted of eight, 1.5 hour classes, and was held at Stone Park Junior High School, Charlottetown, PEI.

Making Your Interface C.R.A.P. Outreach talk presented to the Charlottetown Chapter of Canada's IT Professional Society (CIPS) meeting. January 2013.

2010 Consider Studying Interactive Systems and Computer Science at University. Outreach presentation given to an eleventh-grade computer class, promoting the study of Computer Science and Interactive Systems at university. December 16th, 2010 at Colonel Gray High School, Charlottetown, PE, in Mr. Michael Peterson's class.

### 7.3 CONFERENCE CHARING

| | |
|---|---|
| 2025-27 | General Co-Chair for the ACM CHI PLAY Symposium. Organized and planned the orchestration of the conference to be held in October 2027 in Montreal, Quebec. Octbober 2027. |
| 2026 | Co-organizer of the Atlantic AI Symposium. Created and organized a conference for approximately 350 attendees at UNB, Fredericton, NB. June 3-5, 2026. |
| 2023 | Papers co-chair for the ACM CHI PLAY Symposium. Handled organization of technical program, organization of paper reviewing, review process and associate chairs. Stratford, ON, Canada, October 10-13, 2023. |
| 2022 | Papers co-chair for the ACM CHI PLAY Conference. Handled organization of technical program, organization of paper reviewing, review process and associate chairs. Bremen, Germany, November 2-5, 2022. |
| 2021 | Panel and Masterclass Co-Chair for the ACM CHI PLAY Conference. October 10-14, 2021. |
| 2020 | Panel and Masterclass Co-Chair for the ACM CHI PLAY Conference. Ottawa, Ontario, November 2-4, 2020. |
| 2017 | Technical Program HCI Chair of the 2017 International Conference on Computer Graphics and Interactive Techniques (Graphics Interface 2017). Edmonton, AB. May 2017.<br>2017 Organizing Co-Chair for 2017 Conference for Mathematics, Statistics and Computer Science (Science Atlantic). Fredericton, NB. October 2017. |

### 7.4 CONFERENCE SESSION CHARING

- May 2024: Session Chair for “Reality-Virtuality Continuum: Interaction and Collaboration” at CHI 2024 (Honolulu, Hawaii).
- April 2021: Session Chair for “VR/AR/MR #3” at CHI 2021 (virtual Yokohma, Japan).
- April 2018: Session Chair for "Beyond the Screen" at CHI 2018 (Montreal, Canada).
- May 2017: Session Chair for "Novel Gaming Interfaces" at CHI 2017 (Denver, USA).
- Oct. 2015: Session Chair for Computer Science session at Science Atlantic (Wolfville, NS)
- Oct. 2014: Session Chair for CHI Play 2014 (Toronto) session on Collaboration.

### 7.5 OTHER ACADEMIC ACTIVITIES

N/A

### 7.6 MEDIA COVERAGE, INTERVIEW OR EXPERT OPINION

| | |
|---|---|
| 2025 | “A conversation about artificial intelligence…and what it means for the future of work and education.” CBC Radio One. Maritime Connection with Preston Mulligan. May 11, 2025. Call-in interview program. https://www.cbc.ca/listen/live-radio/1-99-maritime-connection/clip/16 145 616-we-continue-conversation-artificial-intelligence…and-means-future |

2025 "Generative AI can help boost productivity in the workplace, expert says". CBC News. Jan 03, 2025. https://www.cbc.ca/news/canada/new-brunswick/generative-ai-productivity-responsibility-1.7416256

2024 "Fredericton working on an artificial intelligence policy to improve the city's efficiency", Global News. https://globalnews.ca/video/10651201/fredericton-working-on-an-artificial-intelligence-policy-to-improve-the-citys-efficiency
July 29, 2024.

2022 "Holograms, headsets and health care: The future is now" Hannah Rudderham, CBC News. https://www.cbc.ca/news/canada/new-brunswick/holograms-future-technology-1.6664471
Nov 26, 2022.

2019 "UNB Student uses Augmented Reality to Help Amputees" CTV Live at Five, https://atlantic.ctvnews.ca/video?clipId=1634092&jwsource=cl
March 19, 2019.

2018 "Facebook Breach" Maritime Noon, CBC Radio One (to all maritime provinces), 1 hour, radio phone-in-show. March 27, 2018.

2018 "Facebook breach not scaring away users: experts". Michael Staples, The Daily Gleaner (newspaper), Fredericton, New Brunswick. March 23rd, 2018.

2017 "UNB prof seeks to level the playing field in video games for out-of-touch parents" by Joseph Tunney, CBC Website, December 16, 2017.
http://www.cbc.ca/news/canada/new-brunswick/unb-computer-science-video-game-balancing-1.4450416

Radio Interview about player balancing research in the UNB HCI Lab by Joseph Tunney, aired on CBC Radio One in New Brunswick. December 16, 2017.

2016 "UNB's Scott Bateman Helps Humans and Computers Communicate" by Cara Smith, Huddle June 20, 2016. http://huddle.today/scott-bateman/

2015 Interviews on the need for teaching Computer Science in grade school. Interviewed by 9 separate CBC Radio morning show hosts from across Canada. CBC Radio. March 24, 2015.

### 7.7 PARTICIPATION AND/OR SUPERVISION OF PROGRAMMING, INNOVATION OR GAME DEVELOPMENT COMPETITIONS

- Fog Cutter: Field Operation Guidance for Emergency Response Through 5G Enabled Remote Holographic Support. BOOST 5G Innovation Camp Winning Concept Video. https://www.fredericton.ca/en/news/city-hall/boost-5g-innovation-camp-winners-announced-fog-cutter-takes-home-top-prize

### 7.8 EXTERNAL REVIEWING FOR CONFERENCES, PEER-REVIEW WORKSHOPS & JOURNALS

**I am a frequent reviewer for leading HCI Venues including CHI, CHI PLAY, etc. HCI Conferences have since approximately 2017 provided awards for peer reviewers and committee members to provide**

**notations to highly useful or outstanding reviews. I have received 13 such special recognitions to date as an external reviewer.**

- 4 recognitions for CHI 2018 Papers
- 2 recognitions for CHI PLAY 2017 Full Papers
- 2 recognitions for CHI 2020 Papers
- 2 recognitions for CHI 2021 Papers
- 2 recognitions for CHI 2022 Papers
- 1 recognition for CHI PLAY 2024 Full Papers